

INVARIANT TWO-JETS AND EFFECTIVE HYPERBOLICITY FOR COMPLEMENTS OF TWO PLANE CURVES

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ABSTRACT. Let $D = C_1 + C_2 \subset \mathbb{P}^2$ be a simple-normal-crossing union of smooth plane curves of degrees $1 \leq d_1 \leq d_2$. We prove an effective Second Main Theorem on a nonempty Zariski-open set of ordered pairs whenever

$$d_1, d_2 \geq 3, \quad \text{or} \quad d_1 = 2, d_2 \geq 5, \quad \text{or} \quad d_1 = 1, d_2 \geq 8.$$

After intersecting with the standard very-general algebraic-hyperbolicity locus, we obtain Kobayashi hyperbolicity and hyperbolic embedding of the complement. The proof uses logarithmic invariant two-jet differentials, the corrected Demailly–El Goul zero-locus argument, and a two-component adaptation of slanted vector fields with base twist $\mathcal{O}_{\mathbb{P}^2}(3)$. The adapted twist three is uniform in (d_1, d_2) : simultaneous cancellation of the two root equations requires third Taylor differences, but no seventh-order polarization. For a linear component the third-remainder space vanishes, and ten compatible mixed generators replace the generic Taylor package. A numerical phase diagram shows that exact finite computation is needed only for the six boundary pairs $(3, 3), (3, 4), (2, 6), (2, 5), (1, 9), (1, 8)$. Their key vanishings are certified by full-column-rank matrices over finite fields. For two cubics, a strengthened range reduces to twelve theorem-relevant degree–twist cases; two low-weight self-tests bring the computational total to 168 character–eigenvalue blocks. The exact source, certificates, and reproduction instructions form an accompanying computational archive. In the two-cubic case the resulting explicit estimate is

$$T_f(r) \leq 57(N_f^{[1]}(r, C_1) + N_f^{[1]}(r, C_2)) + o(T_f(r)) \parallel.$$

For a conic and a quintic one may take the explicit coefficient 45, and for a line and an octic one may take 69.

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1. INTRODUCTION

1.1. Historical setting. The logarithmic hyperbolicity problem asks how the geometry of a divisor $D \subset \mathbb{P}^n$ constrains entire curves in $\mathbb{P}^n \setminus D$. In dimension one, the model is Picard’s theorem: the complement of three points in $\mathbb{P}^1$ admits no nonconstant entire curve. In higher dimension the guiding principle is that sufficiently positive logarithmic cotangent geometry should force the same rigidity. The Green–Griffiths construction and Demailly’s reparametrization-invariant refinement turn this principle into differential equations on jets of entire curves. In the logarithmic setting this leads to the bundles

$$E_{k,m}T_X^*(\log D),$$

whose negatively twisted sections vanish along every entire curve in the complement; see [2, 4]. The central difficulty is not merely to produce one equation by Riemann–Roch, but to control its base locus well enough to obtain an independent equation.

For logarithmic surfaces, two complementary ideas made this strategy effective. Miyaoka’s zero-locus method (see also Lu–Yau), in the form used by Demailly–El Goul [14, 11, 3], produces a second equation intrinsically on an irreducible component of the first equation’s zero divisor. Siu’s slanted vector-field method [19], developed effectively in dimension two by

Păun [15] and in the logarithmic setting by Rousseau [16], uses the natural action of invariant fields on the universal jet space to turn the first equation ω_1 into a genuine global second jet differential ω_2 . The gain from differentiation is governed by the pole order of the available vector fields; this is why replacing a twist seven construction by twist three has a direct quantitative consequence for a Second Main Theorem.

For the complement of one smooth plane curve, Rousseau obtained the long-standing degree bound $d \geq 14$. Recent work of Hou–Huynh–Merker–Xie lowered it to $d \geq 12$ by combining invariant two-jets with a finite key-vanishing calculation [6]; a subsequent symmetry method sharpened the computational framework further [8]. A related three-component model—three general conics—admits a particularly simple logarithmic frame and yields an effective Second Main Theorem [5]. These developments show that low-degree questions are controlled by a delicate interface between intrinsic jet geometry, explicit slanted fields, and exact finite linear algebra.

The chronological starting point is the three-conics paper [5]. Its decisive innovation is the exponent bound that converts an a priori infinite jet-vanishing problem into finite exact algebra; its Second Main Theorem argument also makes essential use of the geometric structure of the Wronskian. The later one-component paper [6] develops this discovery into a one-component theory, including the local invariant-two-jet normal form and the first concrete implementation of the finite Key Vanishing strategy. The higher-direct-image, zero-locus, and algebraic-extension ingredients are inherited from the preceding work; at each use we identify both the source and the two-component specialization. The symmetry organization follows [8]. The pole-order-three acceleration package is adapted from an announced revision of that preprint; because it is not yet contained in the public version, Section 5 gives the complete two-component argument. The principal new technical result here is the Key Vanishing Lemma for two components: its coupled normal form, unequal-degree divisibility conditions, symmetry decomposition, and exact finite-rank certificates do not occur in any of the three preceding papers. This is the fourth paper of the program, but it is not merely an application of the preceding strategy: the two-component vanishing requires a new coupled proof with two residue directions, degree-sensitive overlap equations, and exact symmetry blocks adapted to ordered pairs. The remaining ingredients require genuine two-component adaptations, although we do not present the overall strategy itself as new.

The case of two components has its own history. Let

$$D = C_1 + C_2 \subset \mathbb{P}^2, \quad 1 \leq d_1 := \deg C_1 \leq d_2 := \deg C_2,$$

where C_1 and C_2 are smooth and transverse. Rousseau’s first result [17, Théorème 1] proves hyperbolicity in the ranges $d_1 \geq 5$, $d_1 = 4, d_2 \geq 7$, $(d_1, d_2) = (4, 4)$, $d_1 = 3, d_2 \geq 9$, and $d_1 = 2, d_2 \geq 12$. His later logarithmic vector-field construction [16, Theorem 3] improves these to $d_1 \geq 4$, $d_1 = 3, d_2 \geq 5$, $d_1 = 2, d_2 \geq 8$, and $d_1 = 1, d_2 \geq 11$. The present paper reaches every pair with $d_1, d_2 \geq 3$, every pair with $d_1 = 2, d_2 \geq 5$, and every pair with $d_1 = 1, d_2 \geq 8$. In particular, it settles the two-cubic and cubic–quartic cases, lowers the degree-two boundary from 8 to 5, and lowers the linear-component boundary from 11 to 8.

Remark 1.1 (The line–septic frontier). For $(d_1, d_2) = (1, e)$ one has

$$\bar{c}_1^2 = (e - 2)^2, \quad \bar{c}_2 = (e - 1)^2, \quad A(1, e) = 4e^2 - 34e + 43.$$

At $e = 10$ this gives

$$A(1, 10) = 103, \quad \rho_{\text{DEG}}(1, 10) = \frac{103}{96} > 1.$$

Thus no finite Key Vanishing Lemma is needed for $e \geq 10$. For $e = 9$ one has $\rho_{\text{DEG}} = 61/84$, leaving only $(m, t) = (3, 3), (4, 3)$; for $e = 8$ one has $\rho_{\text{DEG}} = 3/8$, leaving

$$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3).$$

Lemma 1.2 certifies precisely these eight cases, and the degree-one construction in Section 5 supplies the required pole-three differentiation package. This is why the theorem begins at $e = 8$.

At $e = 7$ one has $A(1, 7) = 1$ and $\rho_{\text{DEG}} = 1/60$; the required minimal profile is

t	m
1	$3 \leq m \leq 60$
2	$61 \leq m \leq 120$
3	$121 \leq m \leq 180$.

Since $A(1, 6) = -17$, the pair $(1, 7)$ is the formal Riemann–Roch endpoint of this two-jet architecture. It should not be confused with a predicted Key Vanishing endpoint: $A(1, 7)$ is barely positive, the list contains 178 minimal cases, and the numerics make it plausible that at least one of the corresponding negatively twisted spaces is nonzero. A future cluster calculation can decide this question, but no claim for $(1, 7)$ is made here.

The two-cubic case is especially close to the conjectural low-degree boundary. Its divisor has only two irreducible components and total degree six. Thus it has the same total degree as the three-conics model but one fewer component, while total degree six is only one above the usual plane Kobayashi benchmark $2 \dim \mathbb{P}^2 + 1 = 5$. The reducible two-component family is of course not the full parameter space of irreducible plane curves of degree six, so these statements are not formally instances of the same genericity conjecture. Nevertheless, simultaneously lowering the number of components and keeping the total degree this small removes much of the logarithmic positivity available in hyperplane- or many-component configurations. In this precise sense, the two-cubic theorem illustrates the low-degree reach of the invariant 2-jet method.

Throughout, a property holds for a *general* ordered pair if it holds on some nonempty Zariski-open subset of the relevant parameter space. It holds for a *very general* ordered pair if its failure locus is contained in a countable union of proper Zariski-closed subsets. We retain the order even when $d_1 = d_2$; no quotient by component exchange is taken.

1.2. What must be adapted for two components. Put

$$s = d_1 + d_2, \quad \kappa = s - 3,$$

and write

$$\overline{T}^* = T_{\mathbb{P}^2}^*(\log D), \quad \overline{T} = T_{\mathbb{P}^2}(-\log D), \quad \overline{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa).$$

We also write

$$h = c_1(\mathcal{O}_{\mathbb{P}^2}(1)), \quad \bar{c}_i = c_i(\overline{T}^*).$$

Unless another space is displayed, $\mathcal{O}(q)$ means $\mathcal{O}_{\mathbb{P}^2}(q)$; for an affine variety U , the notation $\mathcal{O}(U)$ denotes its ring of regular functions. The global Demailly–Semple strategy is inherited from the three works just cited. Its implementation for two components is nevertheless not obtained by applying the one-component formulas twice. We record the necessary adaptations, distinguishing them from the main new key-vanishing argument.

First, the universal root cover now has two logarithmic root variables. A parameter vector field must preserve two first-order tangency equations and two second-order equations, while both equations share the same quadratic acceleration in the base variables. Correcting the field for C_1 alone generally destroys tangency to C_2 . We solve instead a simultaneous pair of Hessian-cancellation equations. Their common solution has third-order Taylor remainder, so after homogenization the field has base twist $\mathcal{O}_{\mathbb{P}^2}(3)$. The construction must also descend through the product of the two cyclic deck groups and remain regular where C_1 and C_2 cross. This compatibility at the double locus has no one-component analogue. When $d_1 = 2$, the quadratic component requires a separate Taylor-remainder argument. When $d_1 = 1$, the third-remainder space is zero; ten explicit line-compatible generators replace it and make the coefficient–root gauge and the double-locus descent transparent. These two borderline analyses are what permit respectively $d_2 \geq 5$ and $d_2 \geq 8$.

Second, the numerical problem is genuinely two-dimensional. The logarithmic Chern numbers depend separately on d_1 and d_2 , and the competition between one-jets, invariant two-jets, and the zero-locus method produces a phase diagram in the (d_1, d_2) -plane. The one-jet Segre number is

$$\bar{c}_1^2 - \bar{c}_2 = d_1 d_2 - 3(d_1 + d_2) + 6.$$

Thus one-jets already treat $d_1 \geq 5$ and $d_1 = 4, d_2 \geq 7$, but they do not see the low-degree corner addressed here. Invariant two-jets reduce that corner to six boundary pairs $(3, 3), (3, 4), (2, 6), (2, 5), (1, 9), (1, 8)$.

Third, the key vanishing calculation is coupled. For one component there is one residue direction and one defining polynomial. Here chart changes mix two logarithmic residues, and unequal degrees carry different homogeneities. We derive a finite normal form that records these coupled transition and divisibility conditions as sparse integral matrices. Equal-degree symmetry splits the two-cubic matrices into character and involution blocks; the unequal-degree pairs require separate normal forms and carefully chosen SNC test divisors. Exact full-column rank modulo one good prime proves rank over $\mathbb{Q}$, and semicontinuity then gives vanishing on a parameter open. This is an exact proof certificate, not a numerical floating-point test. The implementation is deliberately workstation-scale: symmetry blocks, sparse on-demand columns, modular elimination, and bounded parallelism keep the certified finite-vanishing range used for SMT-57 within a laptop-class configuration with 32 GB of memory and 1 TB of storage. Deterministic hashes and independent shards preserve reproducibility when the same computation is moved to a larger machine. This differs from the extreme residual systems in the third paper of the series, whose intended full computations require high-memory server or cluster resources; here the required range is engineered to remain within laptop-scale hardware.

The principal technical originality of the paper is the third point: the coupled finite key-vanishing theorem and its proof certificates. The first point is a nontrivial two-component adaptation of the $\mathcal{O}(3)$ mechanism of [8], and the second is the numerical bookkeeping needed to deploy the inherited strategy in the (d_1, d_2) -plane. Together these ingredients retain pole order three in the two-component setting and make the new degree range possible.

1.3. The finite Key Vanishing Lemma. For $d \geq 1$ write $V_d = H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d))$. For fixed degrees put

$$\mathcal{S}_{d_1, d_2} = \mathbb{P}(V_{d_1}) \times \mathbb{P}(V_{d_2}),$$

and denote by $\mathcal{S}_{d_1, d_2}^{\text{snc}}$ its nonempty open simple-normal-crossing locus. Define

$$A(d_1, d_2) = 4(d_1^2 + d_2^2) + 17d_1 d_2 - 51(d_1 + d_2) + 90,$$

$$\rho_{\text{DEG}}(d_1, d_2) = \frac{A(d_1, d_2)}{12(d_1 + d_2 - 3)}.$$

For a genuine second-level divisor only $m \geq 3$ occurs. For the five boundary pairs other than $(3, 3)$ set

$$\mathfrak{F}_{\text{bd}}(d_1, d_2) = \left\{ (m, t) \in \mathbb{Z}_{\geq 3} \times \{1, 2, 3\} : \frac{t}{m} \geq \rho_{\text{DEG}}(d_1, d_2) \right\}.$$

For the conic–quintic pair we certify one additional minimal case and write

$$\mathfrak{F}_{45}(2, 5) = \left\{ (m, t) \in \mathbb{Z}_{\geq 3} \times \{1, 2, 3\} : \frac{t}{m} > \frac{1}{3} \right\} = \mathfrak{F}_{\text{bd}}(2, 5) \cup \{(8, 3)\}.$$

For two cubics we use the stronger range

$$\mathfrak{F}_{57}(3, 3) = \{(m, t) \in \mathbb{Z}_{\geq 1} \times \{1, 2, 3\} : 1 \leq m \leq 5t - 1\}.$$

The two cases with $m = 1, 2$ are included as low-weight consistency tests; the geometric argument uses only the twelve cases with $m \geq 3$.

Lemma 1.2 (Finite key vanishing). *For each of the six degree pairs*

$$(d_1, d_2) \in \{(3, 3), (3, 4), (2, 6), (2, 5), (1, 9), (1, 8)\}$$

there exists a nonempty Zariski-open subset $\mathcal{U}_{\text{van}}(d_1, d_2) \subset \mathcal{S}_{d_1, d_2}^{\text{snc}}$ such that, for every parameter a in this subset, one has

$$(1.1) \quad H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)) = 0$$

for every $(m, t) \in \mathfrak{F}_{57}(3, 3)$ when $(d_1, d_2) = (3, 3)$, for every $(m, t) \in \mathfrak{F}_{45}(2, 5)$ when $(d_1, d_2) = (2, 5)$, and for every $(m, t) \in \mathfrak{F}_{\text{bd}}(d_1, d_2)$ for the other four pairs.

The finite sets, after reduction to the least twist for each m , are shown below. The second column is the zero-locus threshold $\rho_{\text{DEG}}(d_1, d_2)$, and the third column lists the pairs (m, t) for which the vanishing (1.1) must be checked. Thus the cubic row records fourteen computations: the four pairs $(m, 1)$ with $1 \leq m \leq 4$, the five pairs $(m, 2)$ with $5 \leq m \leq 9$, and the five pairs $(m, 3)$ with $10 \leq m \leq 14$; its first two entries are the self-tests just described.

(d_1, d_2)	ρ_{DEG}	minimal pairs (m, t)
$(3, 3)$	1/4	$1 \leq m \leq 4$ for $t = 1$; $5 \leq m \leq 9$ for $t = 2$; $10 \leq m \leq 14$ for $t = 3$
$(3, 4)$	37/48	$(3, 3)$
$(2, 6)$	23/30	$(3, 3)$
$(2, 5)$	19/48	$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3)$
$(1, 9)$	61/84	$(3, 3), (4, 3)$
$(1, 8)$	3/8	$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3)$

Multiplication by a nonzero homogeneous polynomial handles all larger twists with the same m . Exact modular full rank at one SNC pair implies rank over $\mathbb{Q}$, hence over $\mathbb{C}$; upper semicontinuity then promotes the vanishing to the asserted parameter open. The complete source code, machine-readable certificates, run logs, hashes, and reproduction instructions form the accompanying computational archive and will also be mirrored on Song-Yan Xie's homepage.¹

1.4. Main statements.

Theorem 1.3 (Effective Second Main Theorem). *Let $1 \leq d_1 \leq d_2$ be integers. Assume that*

$$d_1, d_2 \geq 3, \quad \text{or} \quad d_1 = 2, d_2 \geq 5, \quad \text{or} \quad d_1 = 1, d_2 \geq 8.$$

There exist a nonempty Zariski-open subset $\mathcal{U}_{\text{SMT}}(d_1, d_2) \subset \mathcal{S}_{d_1, d_2}^{\text{snc}}$ and an effectively computable constant $\mathcal{A}_{d_1, d_2} > 0$, both depending only on the ordered degree pair. For every a in this subset, write $D_a = C_{1,a} + C_{2,a}$. Then every algebraically nondegenerate entire curve $f : \mathbb{C} \rightarrow \mathbb{P}^2$ whose image is not contained in D_a satisfies

$$(1.2) \quad T_f(r) \leq \mathcal{A}_{d_1, d_2} N_f^{[1]}(r, D_a) + o(T_f(r)) \parallel,$$

where, throughout the paper, truncation for a reduced divisor is taken componentwise:

$$N_f^{[1]}(r, D_a) = N_f^{[1]}(r, C_{1,a}) + N_f^{[1]}(r, C_{2,a}).$$

For $(d_1, d_2) = (3, 3)$ one may take $\mathcal{A}_{3,3} = 57$; for $(d_1, d_2) = (2, 5)$ one may take $\mathcal{A}_{2,5} = 45$; and for $(d_1, d_2) = (1, 8)$ one may take $\mathcal{A}_{1,8} = 69$.

Let $\mathcal{S}_{\text{Chen}}(d_1, d_2)$ denote the complement of the countable union of proper closed subsets excluded by Theorem 2.13.

Corollary 1.4 (Hyperbolicity and hyperbolic embedding). *Under the degree assumptions of Theorem 1.3, the complement $\mathbb{P}^2 \setminus D_a$ is Kobayashi hyperbolic and hyperbolically embedded in $\mathbb{P}^2$ for every $a \in \mathcal{U}_{\text{SMT}}(d_1, d_2) \cap \mathcal{S}_{\text{Chen}}(d_1, d_2)$. In particular, this conclusion holds for a very general ordered pair of smooth transverse curves in the stated degree range.*

¹Computational evidence: <https://xiesongyan.github.io/>.

The two genericity conditions enter at different stages. The Second Main Theorem holds on the nonempty Zariski-open set $\mathcal{U}_{\text{SMT}}$, whereas Xi Chen's theorem excludes a countable union $\bigcup_{\nu \geq 1} Z_\nu$ of proper closed subsets. Since $\mathcal{S}_{d_1, d_2}^{\text{snc}}$ is irreducible, the complement of $\mathcal{U}_{\text{SMT}}$ is itself a proper closed subset. Hence

$$\mathcal{U}_{\text{SMT}} \cap \mathcal{S}_{\text{Chen}} = \mathcal{S}_{d_1, d_2}^{\text{snc}} \setminus \left((\mathcal{S}_{d_1, d_2}^{\text{snc}} \setminus \mathcal{U}_{\text{SMT}}) \cup \bigcup_{\nu \geq 1} Z_\nu \right)$$

is a nonempty very-general subset. Thus the open condition needed for the effective estimate and the very-general condition needed to exclude algebraically degenerate curves remain logically separate.

1.5. Proof strategy and computational frontier. Riemann–Roch and logarithmic H^2 vanishing first produce a negatively twisted invariant two-jet differential. The Demailly–El Goul reduction then selects an irreducible factor ω_1 of weighted degree m and twist $-t$, without decreasing the twist-to-weight ratio. If $f^*\omega_1 \neq 0$, the jet Second Main Theorem gives the estimate directly. A zero component pulled back from the first Semple level is handled by McQuillan's theorem. For a genuine second-level component, $t > 3$ is handled by differentiating with the mixed $\mathcal{O}(3)$ fields. If $t \leq 3$, the numerical threshold either places t/m in the zero-locus range or reduces the problem to the finite set in Lemma 1.2; the restricted-bigness argument then supplies a second section on the zero divisor. Finally, Xi Chen's inequality excludes algebraically degenerate entire curves from a very general complement.

The explicit endpoint constants use sharper versions of this dichotomy. For a conic and a quintic, the additional certified case $(m, t) = (8, 3)$ moves the split to $\rho = t/m = 1/3$. The branch $\rho \leq 1/3$ has zero-locus coefficient strictly smaller than 45, whereas $\rho > 1/3$ forces $t \geq 4$ and $m \leq 3t - 1$, giving differentiated coefficient at most 11. Choosing the initial ratio above $1/10$ bounds the direct coefficient strictly below 10.

For two cubics the strengthened finite vanishing range moves the split to $\rho = 1/5$. When $\rho \leq 1/5$, the limiting zero-locus coefficient is

$$\frac{57 + \sqrt{3189}}{2} < 57.$$

Thus the coefficient in the zero-locus estimate can be chosen below 57. When $\rho > 1/5$, the finite theorem excludes $t \leq 3$; hence $t \geq 4$, and integrality gives $m \leq 5t - 1$. The $\mathcal{O}(3)$ derivative therefore gives

$$\frac{m}{t-3} \leq \frac{5t-1}{t-3} \leq 19.$$

Together with the direct and McQuillan branches, this proves the stated integer constant 57.

Section 9 explains why we retain these convenient, non-optimal coefficients and identifies the precise additional Key Vanishing Lemmas needed to push the same method to its numerical limit.

Section 2 collects the logarithmic-jet notation and external analytic inputs. Section 3 computes the logarithmic Chern numbers and the two numerical thresholds. Section 4 constructs the first differential. Section 5 proves the simultaneous mixed $\mathcal{O}(3)$ construction, which is the geometric core of the paper. Section 6 combines the zero-locus and differentiation mechanisms, and Section 7 assembles the main results. Section 8 proves the finite Key Vanishing Lemma by symmetry-adapted exact linear algebra; the complete C++ source and certificate protocol are distributed in the supplementary archive. The final Section 9 describes the remaining computational frontier for the best constants obtainable from the present geometry.

2. LOGARITHMIC JETS AND EXTERNAL INPUTS

2.1. Nevanlinna functions. Let ω_{FS} be the Fubini–Study form. We use the standard Nevanlinna functions and normalization; see [18]. For $f : \mathbb{C} \rightarrow \mathbb{P}^2$, set

$$T_f(r) = \int_1^r \frac{ds}{s} \int_{|z| < s} f^* \omega_{\text{FS}}.$$

If $E \subset \mathbb{P}^2$ is an irreducible divisor and $f(\mathbb{C}) \not\subset E$, let

$$N_f^{[1]}(r, E) = \int_1^r \sum_{|z| < s} \min\{1, \text{ord}_z f^* E\} \frac{ds}{s}.$$

For the reduced divisor $D = C_1 + C_2$, we use the componentwise convention

$$N_f^{[1]}(r, D) := N_f^{[1]}(r, C_1) + N_f^{[1]}(r, C_2).$$

The symbol $\parallel$ means that the inequality holds for all $r > 1$ outside a set of finite Lebesgue measure. For an algebraically nondegenerate curve in $\mathbb{P}^2$, the standard growth lemma gives

$$O(\log T_f(r) + \log r) = o(T_f(r)) \parallel,$$

so we shall replace the logarithmic error term by $o(T_f)$ after applying a jet-differential estimate.

2.2. Invariant logarithmic two-jets. We follow the standard invariant logarithmic jet notation of [2, 4]. Locally on a logarithmic surface, an invariant two-jet differential of weighted degree m has the form

$$(2.1) \quad \sum_{0 \leq j \leq \lfloor m/3 \rfloor} \sum_{\alpha_1 + \alpha_2 = m - 3j} R_{\alpha_1, \alpha_2, j} (f'_1)^{\alpha_1} (f'_2)^{\alpha_2} (f'_1 f'_2 - f'_2 f'_1)^j.$$

Its associated graded bundle is

$$(2.2) \quad \text{Gr}^\bullet E_{2,m} \bar{T}^* = \bigoplus_{0 \leq j \leq \lfloor m/3 \rfloor} S^{m-3j} \bar{T}^* \otimes \bar{K}^j.$$

2.3. The logarithmic Demailly–Semple tower. Set $X_0 = \mathbb{P}^2$ and $V_0 = \bar{T}$. We use the convention that $\mathbb{P}(V)$ parametrizes lines in V , so that $\mathcal{O}_{\mathbb{P}(V)}(-1)$ is the tautological subbundle. Let

$$X_1 = \mathbb{P}(V_0), \quad X_2 = \mathbb{P}(V_1),$$

be the first two levels of the logarithmic Demailly–Semple tower, with projections $\pi_{2,1} : X_2 \rightarrow X_1$ and $\pi_{2,0} : X_2 \rightarrow X_0$. Write

$$u_1 = c_1(\pi_{2,1}^* \mathcal{O}_{X_1}(1)), \quad u_2 = c_1(\mathcal{O}_{X_2}(1)), \quad h = \pi_{2,0}^* c_1(\mathcal{O}_{\mathbb{P}^2}(1)),$$

and

$$\mathcal{O}_{X_2}(a, b) := \pi_{2,1}^* \mathcal{O}_{X_1}(a) \otimes \mathcal{O}_{X_2}(b).$$

Let $\Gamma_2 \subset X_2$ be the vertical divisor at the second stage. With the above convention,

$$\mathcal{O}_{X_2}(\Gamma_2) \simeq \mathcal{O}_{X_2}(-1, 1), \quad \mathcal{O}_{X_2}(3) \otimes \mathcal{O}_{X_2}(-2\Gamma_2) \simeq \mathcal{O}_{X_2}(2, 1).$$

Let $\mathcal{O}_{X_2}(m)$ denote the standard level-two Demailly–Semple tautological bundle of weighted degree m . The direct-image formula identifies invariant two-jets with its sections:

$$(2.3) \quad (\pi_{2,0})_* \mathcal{O}_{X_2}(m) \simeq E_{2,m} \bar{T}^*.$$

We shall also use the following more precise direct-image statement for the bundle $\mathcal{O}_{X_2}(2, 1)$. Its two-stage filtration proof is the one used in [5, Proposition 6.1]. Keeping every negative-degree graded piece gives the sharp criterion in terms of the single defect $a_1 - 2a_2$; we record the resulting statement and proof for later use.

Proposition 2.1 (Sharp higher direct images). *Let $a_1, a_2 \in \mathbb{Z}$. If $a_2 = -1$, then*

$$R^q(\pi_{2,0})_* \mathcal{O}_{X_2}(a_1, -1) = 0 \quad (q \geq 0).$$

Suppose that $a_2 \geq 0$ and put $d = a_1 - 2a_2$. Then

$$R^q(\pi_{2,0})_* \mathcal{O}_{X_2}(a_1, a_2) = 0 \quad (q \geq 2),$$

and

$$R^1(\pi_{2,0})_* \mathcal{O}_{X_2}(a_1, a_2) = 0 \iff d \geq -1.$$

If $d \leq -2$, the nonzero R^1 has a finite filtration with successive quotients

$$(2.4) \quad \overline{K}^{a_2-k-1} \otimes \text{Sym}^{-d-3k-2} \overline{T}, \quad 0 \leq k \leq \min \left\{ a_2, \left\lfloor \frac{-d-2}{3} \right\rfloor \right\}.$$

Proof. Write $q = \pi_{2,1}$ and $\pi = \pi_{1,0}$. If $a_2 = -1$, the restriction to each q -fiber is $\mathcal{O}_{\mathbb{P}^1}(-1)$, so every direct image under q , and hence under $\pi_{2,0}$, vanishes. Assume $a_2 \geq 0$. The projective-bundle formula gives

$$R^j q_* \mathcal{O}_{X_2}(a_1, a_2) = 0 \quad (j > 0), \quad q_* \mathcal{O}_{X_2}(a_1, a_2) = \mathcal{O}_{X_1}(a_1) \otimes \text{Sym}^{a_2} V_1^*.$$

The dual directed sequence

$$0 \longrightarrow \mathcal{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi^* \overline{K} \otimes \mathcal{O}_{X_1}(-2) \longrightarrow 0$$

induces a filtration whose graded quotients are

$$(2.5) \quad \pi^* \overline{K}^{a_2-k} \otimes \mathcal{O}_{X_1}(d+3k), \quad 0 \leq k \leq a_2.$$

For the rank-two projective bundle of lines, $R^1 \pi_* \mathcal{O}_{X_1}(n) = 0$ for $n \geq -1$, while relative Serre duality gives, for $n \leq -2$,

$$R^1 \pi_* \mathcal{O}_{X_1}(n) \simeq \text{Sym}^{-n-2} \overline{T} \otimes \det \overline{T} = \text{Sym}^{-n-2} \overline{T} \otimes \overline{K}^{-1}.$$

Applying this to (2.5) gives precisely the quotients (2.4). In the bad range $d+3k \leq -2$, the corresponding ordinary direct image is zero; ordering the filtration from the good range to the bad range shows through the long exact sequences that these R^1 terms assemble by successive extensions. There are no direct images in degree at least two because both stages have relative dimension one and the first push has no positive higher direct images. The vanishing criterion is therefore exactly $d \geq -1$. $\square$

Remark 2.2 (Fiberwise interpretation). On a fiber $F \simeq \mathbb{F}_3$ of $\pi_{2,0}$,

$$\mathcal{O}_{X_2}(a_1, a_2)|_F \simeq \mathcal{O}_{\mathbb{F}_3}(a_2 E + (a_1 + a_2)f),$$

and for $a_2 \geq 0$ its cohomology is controlled by the line-bundle degrees $d+3k$, $0 \leq k \leq a_2$. Thus the defect $a_1 - 2a_2$ is sharp; conditions involving a_1 and a_2 separately do not suffice.

Lemma 2.3 (Direct images of $\mathcal{O}_{X_2}(2p, p)$). *For every integer $p \geq 1$ one has*

$$(2.6) \quad (\pi_{2,0})_* \mathcal{O}_{X_2}(2p, p) \simeq E_{2,3p} \overline{T}^*.$$

Moreover,

$$(2.7) \quad R^q(\pi_{2,0})_* \mathcal{O}_{X_2}(2p, p) = 0 \quad (q \geq 1).$$

Proof. The identification (2.6) is the standard Demailly–Semple direct-image formula for the weight- $3p$ subbundle; see [3, Lemma 3.3(c)]. The higher direct images vanish by Proposition 2.1, because $a_2 = p \geq 0$ and $a_1 - 2a_2 = 2p - 2p = 0$. $\square$

Thus the p -th power of $\mathcal{O}_{X_2}(2, 1)$ corresponds canonically to weighted degree $3p$.

We shall repeatedly use the following elementary fact about lifted curves.

Lemma 2.4 (The regular lift avoids the vertical divisor). *Let $f : \mathbb{C} \rightarrow X_0$ be nonconstant and satisfy $f(\mathbb{C}) \not\subset D$. Then its logarithmic lift $f_{[2]}$ is well defined after holomorphic extension across the isolated zeros of f' , and*

$$f_{[2]}(\mathbb{C}) \not\subset \Gamma_2.$$

Proof. On the nonempty open set on which $f' \neq 0$, the first lift is $f_{[1]}(z) = (f(z), [f'(z)])$. Its derivative is tangent to V_1 , and its projection by $d\pi_{1,0}$ is $f'(z) \neq 0$. Hence the line $[f'_{[1]}(z)] \subset V_{1,f_{[1]}(z)}$ is not vertical for $\pi_{1,0}$. By definition, Γ_2 consists precisely of the vertical lines at the second stage, so $f_{[2]}(z) \notin \Gamma_2$ on this open set. The standard removable-singularity construction for Semple lifts extends $f_{[1]}$ and $f_{[2]}$ through the isolated zeros of f' . Since the complement of Γ_2 has already met the lifted curve, the whole image cannot be contained in Γ_2 . $\square$

2.4. Jet Second Main Theorem and McQuillan's theorem. We use the following standard jet-differential estimate; compare [2] and [5, Theorem 1.2]. The invariant logarithmic jet bundle $E_{k,m}T_{\mathbb{P}^2}^*(\log D)$ is naturally a subbundle of the Green–Griffiths bundle appearing in the statement, so the estimate applies to the invariant differentials constructed below.

Theorem 2.5 (Jet differential Second Main Theorem). *Let $k, m, t \geq 1$ be integers. If*

$$0 \neq \omega \in H^0(\mathbb{P}^2, E_{k,m}^{\text{GG}}T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}(-t))$$

and $f : \mathbb{C} \rightarrow \mathbb{P}^2$ satisfies $f(\mathbb{C}) \not\subset D$ and $f^\omega \neq 0$, then*

$$(2.8) \quad T_f(r) \leq \frac{m}{t} N_f^{[1]}(r, D) + O(\log T_f(r) + \log r) \parallel.$$

We also use McQuillan's algebraic-degeneracy theorem in the following quantitative form; see [12] and [5, Theorem 1.1].

Theorem 2.6 (McQuillan). *Let $f : \mathbb{C} \rightarrow \mathbb{P}^2$ be algebraically nondegenerate. If its first logarithmic lift is algebraically degenerate, then for a normal-crossing divisor of total degree $d \geq 4$,*

$$(d-3)T_f(r) \leq N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

For $d = s = d_1 + d_2$ this gives

$$(2.9) \quad T_f(r) \leq \frac{1}{s-3} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

2.5. Cohomological and zero-locus inputs. We next record the vanishing and weighted-degree estimates that produce the negatively twisted differentials used in the two main cases.

Lemma 2.7 (Logarithmic Bogomolov vanishing). *Let $D = C_1 + C_2$ be a transverse pair of smooth plane curves, put $\kappa = d_1 + d_2 - 3 > 0$, $\bar{K} = K_{\mathbb{P}^2} + D$, and set $\bar{T} = T_{\mathbb{P}^2}(-\log D)$. For integers $p \geq 0$ and q ,*

$$H^0(\mathbb{P}^2, S^p \bar{T} \otimes \bar{K}^q) = 0 \quad \text{if} \quad p > 2q.$$

Proof. This is the logarithmic Bogomolov-type vanishing theorem [5, Theorem 2.1]. $\square$

Theorem 2.8 (The required logarithmic H^2 vanishing). *Let $D = C_1 + C_2$ be a transverse pair of smooth plane curves with $\kappa = d_1 + d_2 - 3 > 0$, and put $\bar{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa)$. For every rational number $0 \leq \delta < 1/3$ and every sufficiently divisible $m \gg 1$,*

$$(2.10) \quad H^2(\mathbb{P}^2, E_{2,m}T_{\mathbb{P}^2}^*(\log D) \otimes \bar{K}^{-\delta m}) = 0.$$

Proof. By Serre duality, the dual of the group in (2.10) is

$$H^0(\mathbb{P}^2, K_{\mathbb{P}^2} \otimes (E_{2,m}\bar{T}^*)^\vee \otimes \bar{K}^{\delta m}).$$

The dual of the invariant-two-jet filtration has graded pieces

$$S^{m-3j}\bar{T} \otimes \bar{K}^{-j} \otimes K_{\mathbb{P}^2} \otimes \bar{K}^{\delta m}, \quad 0 \leq j \leq \lfloor m/3 \rfloor.$$

Because m is sufficiently divisible, δm is an integer. Multiplication by a nonzero section of $\mathcal{O}_{\mathbb{P}^2}(3)$ injects each graded piece into

$$(2.11) \quad S^{m-3j}\bar{T} \otimes \bar{K}^{\delta m-j}.$$

For (2.11), the inequality in Lemma 2.7 is equivalent to

$$(m - 3j) - 2(\delta m - j) = (1 - 2\delta)m - j \geq (2/3 - 2\delta)m > 0.$$

Thus every graded piece has vanishing H^0 . Induction along the filtration gives the vanishing of the whole Serre-dual group, proving (2.10). $\square$

For the zero-locus argument, write

$$\bar{c}_i = c_i(T_{\mathbb{P}^2}^*(\log D)), \quad h = c_1(\mathcal{O}_{\mathbb{P}^2}(1)),$$

and identify these classes with their pullbacks to the Semple tower. For $\rho = t/m$, set

$$(2.12) \quad Q_{\text{DEG}}(\rho, \tau) = 3\tau^2 + (9\rho - 12(\bar{c}_1 \cdot h))\tau + (13\bar{c}_1^2 - 9\bar{c}_2) - 12(\bar{c}_1 \cdot h)\rho.$$

When its value at $\tau = 0$ is positive, denote its smaller positive root by

$$(2.13) \quad \tau_1(\rho) = \frac{12\kappa - 9\rho - \sqrt{(12\kappa - 9\rho)^2 - 12((13\bar{c}_1^2 - 9\bar{c}_2) - 12\kappa\rho)}}{6}.$$

Section 3 derives these expressions from the two-component Chern classes and the tower intersection relations.

Lemma 2.9 (Tautological evaluation of a restricted section). *Let $Z \subset X_2$ be an effective Cartier divisor and let*

$$\sigma \in H^0(Z, \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t})|_Z), \quad p, \tilde{t} \in \mathbb{Z}_{>0}.$$

Let $f: \mathbb{C} \rightarrow \mathbb{P}^2$ be nonconstant, assume $f(\mathbb{C}) \not\subset D$, and suppose that its regular logarithmic second lift is contained in Z . On $\mathbb{C} \setminus f^{-1}(D)$, regard $df_{[1]}: T_{\mathbb{C}} \rightarrow f_{[1]}^ V_1$ as a differential in the directed structure. Wherever it is nonzero, its image is the tautological line $f_{[2]}^* \mathcal{O}_{X_2}(-1)$. It therefore defines, by meromorphic continuation across the remaining discrete set, the intrinsic tautological derivative*

$$\mathcal{D}_D f_{[1]} \in H_{\text{mer}}^0(\mathbb{C}, K_{\mathbb{C}} \otimes f_{[2]}^* \mathcal{O}_{X_2}(-1)).$$

Then the canonical inclusion

$$\iota_p: \mathcal{O}_{X_2}(2p, p) = \mathcal{O}_{X_2}(3p)(-2p\Gamma_2) \hookrightarrow \mathcal{O}_{X_2}(3p)$$

defines the intrinsic meromorphic evaluation

$$(2.14) \quad s_{\sigma, f} := \left\langle f_{[2]}^*(\iota_p \sigma), (\mathcal{D}_D f_{[1]})^{\otimes 3p} \right\rangle \in H_{\text{mer}}^0(\mathbb{C}, K_{\mathbb{C}}^{3p} \otimes f^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t})).$$

This construction does not require an ambient lift of σ . Moreover,

$$(2.15) \quad \text{div}_{\infty}(s_{\sigma, f}) \leq 3p \sum_{i=1}^2 (f^* C_i)_{\text{red}},$$

and

$$(2.16) \quad \int_0^{2\pi} \log^+ \|s_{\sigma, f}(re^{\sqrt{-1}\theta})\| \frac{d\theta}{2\pi} = O(\log^+ T_f(r) + \log r) \quad \parallel.$$

At an isolated zero of f' outside $f^{-1}(D)$, the evaluation has only a removable singularity or a zero. At a zero lying over D , no pole occurs beyond the logarithmic boundary poles already counted in (2.15).

Proof. Pairing its $3p$ -th tensor power with $f_{[2]}^*(\iota_p \sigma)$ gives (2.14). Since this is a pairing of dual tautological lines, the result is independent of all Semple coordinates and all local representatives of σ .

Equip the line bundles with smooth Hermitian metrics. Since Z is projective, the norm of the holomorphic section $\iota_p \sigma$ is bounded on Z . Hence

$$\|s_{\sigma, f}\| \leq C \|\mathcal{D}_D f_{[1]}\|^{3p}$$

for a uniform constant C .

In coordinates in which $D = \{x_1 \cdots x_q = 0\}$, the horizontal boundary components of the derivative of the regular first lift are $(x_i \circ f)' / (x_i \circ f)$, while the vertical projective-coordinate components are holomorphic. Thus the tautological derivative has at most a simple pole along each reduced pullback $(f^*C_i)_{\text{red}}$.

By Lemma 2.4, the Semple lift extends across the isolated zeros of f' . At a critical point outside $f^{-1}(D)$, the projective ratios extend and the evaluation has only a removable singularity or a zero. At a critical point over D , a logarithmic component may retain a simple pole; that pole is already counted by the reduced boundary pullback, and no additional Semple-type pole appears. This proves (2.15). Choose a finite logarithmic base cover and, above it, finitely many relatively compact Semple charts. A partition of unity is used only to estimate the norm of this single global section. On each chart, the components of $\mathcal{D}_D f_{[1]}$ are logarithmic derivatives of local coordinate functions. A projective Semple coordinate u is a rational function on X_1 and, in a logarithmic frame, a ratio of first logarithmic derivatives. Hence

$$T_{u \circ f_{[1]}}(r) = O(T_f(r) + \log r).$$

For a partition function χ supported in a relatively compact Semple chart, choose $a \notin \overline{u(\text{supp } \chi)}$. On that support, $u - a$ is bounded above and away from zero, and

$$(u \circ f_{[1]})' = (u \circ f_{[1]} - a) \frac{(u \circ f_{[1]})'}{u \circ f_{[1]} - a}.$$

The logarithmic-derivative lemma applied to $u \circ f_{[1]} - a$ controls the vertical component by

$$O(\log^+ T_f(r) + \log r).$$

Summing over the finite cover, together with the analogous horizontal estimates, gives (2.16). This is the argument of [7, Thm. 3.1, proof of (3.3)–(3.6)], with the Semple-coordinate adaptation made explicit. $\square$

Lemma 2.10 (Second Main Theorem for a section restricted to Z). *Let $Z \subset X_2$ be an effective Cartier divisor and let*

$$\sigma \in H^0(Z, \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t})|_Z), \quad p, \tilde{t} \in \mathbb{Z}_{>0}.$$

Suppose that $f : \mathbb{C} \rightarrow \mathbb{P}^2$ is nonconstant, $f(\mathbb{C}) \not\subset D$, $f_{[2]}(\mathbb{C}) \subset Z$, and the canonical evaluation $s_{\sigma,f}$ of Lemma 2.9 is not identically zero. Then

$$(2.17) \quad \tilde{t} T_f(r) \leq 3p \sum_{i=1}^2 N_f^{[1]}(r, C_i) + O(\log T_f(r) + \log r) \quad \parallel.$$

In particular, if f is algebraically nondegenerate, then

$$T_f(r) \leq \frac{3p}{\tilde{t}} N_f^{[1]}(r, D) + o(T_f(r)) \quad \parallel.$$

Proof. By Lemma 2.9, the section $s_{\sigma,f}$ is globally and intrinsically defined, its pole divisor is bounded by (2.15), and its proximity term satisfies (2.16). Use the standard coordinate z on $\mathbb{C}$ to trivialize $K_{\mathbb{C}}^{3p}$ by the global frame $(dz)^{3p}$ and give it the flat metric. Equip $\mathcal{O}_{\mathbb{P}^2}(-1)$ with the metric dual to the Fubini–Study metric. Poincaré–Lelong and Jensen’s formula give

$$\begin{aligned} \tilde{t} T_f(r) &\leq 3p \sum_{i=1}^2 N_f^{[1]}(r, C_i) \\ &\quad + O(\log^+ T_f(r) + \log r) \quad \parallel. \end{aligned}$$

This is (2.17). It is the intrinsic restricted-section version of the argument in [5, Proposition 6.3 and Theorem 6.4]; the analytic norm estimate is the one in [7, Theorem 3.1, proof of (3.3)–(3.6)]. $\square$

Theorem 2.11 (Two-component zero-locus theorem and standard estimate). *Let ω_1 be an irreducible nonzero section of*

$$E_{2,m}\overline{T}^* \otimes \mathcal{O}(-t), \quad \rho = t/m,$$

whose zero divisor on X_2 is $Z + b\Gamma_2$, where Z is irreducible and reduced and

$$(2.18) \quad Z \sim b_1 u_1 + b_2 u_2 - th, \quad b_1 \geq 2b_2 > 0, \quad b_1 + b_2 = m.$$

Suppose

$$0 < \rho < \rho_{\text{DEG}} := \frac{13\bar{c}_1^2 - 9\bar{c}_2}{12(\bar{c}_1 \cdot h)}.$$

Assume $\kappa = d_1 + d_2 - 3 \geq 3$. Let $\tau_1(\rho)$ be the smaller positive root of $Q_{\text{DEG}}(\rho, \tau) = 0$, explicitly given in (2.13). For every rational

$$0 < \tau < \min\{3, \tau_1(\rho)\},$$

the $\mathbb{Q}$ -line bundle

$$(\mathcal{O}_{X_2}(2, 1) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tau))|_Z$$

is big. Moreover, if $f : \mathbb{C} \rightarrow \mathbb{P}^2$ is algebraically nondegenerate and $f^ \omega_1 \equiv 0$, then either*

$$(2.19) \quad T_f(r) \leq \frac{3}{\tau} N_f^{[1]}(r, D) + o(T_f(r)) \parallel,$$

or the first logarithmic lift $f_{[1]}$ is algebraically degenerate.

Proof. The restricted-bigness calculation follows the method of [5, Proposition 6.2], with the two-component intersection polynomial computed below. The local frame and analytic estimate are adapted from [5, Proposition 6.3 and Theorem 6.4] and are recorded in Lemma 2.10.

Put

$$L_\tau = \mathcal{O}_{X_2}(2, 1) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tau).$$

The intersection computation on the logarithmic Demailly–Semple tower gives

$$(2.20) \quad c_1(L_\tau)^3 \cdot Z = mQ_{\text{DEG}}(\rho, \tau).$$

By the definition of $\tau_1(\rho)$, the right-hand side is positive in the stated range.

We prove bigness without inferring it from (2.20) alone. Let $p \gg 1$ be sufficiently divisible, so that $p\tau/\kappa \in \mathbb{Z}$ (and hence $p\tau \in \mathbb{Z}$) and L_τ^p is an honest line bundle. Since X_2 is smooth and Z is an effective Cartier divisor, the exact sequence

$$0 \rightarrow L_\tau^p(-Z) \rightarrow L_\tau^p \rightarrow L_\tau^p|_Z \rightarrow 0$$

and Hirzebruch–Riemann–Roch on the smooth fourfold X_2 give

$$(2.21) \quad \chi(Z, L_\tau^p|_Z) = \chi(X_2, L_\tau^p) - \chi(X_2, L_\tau^p(-Z))$$

$$(2.22) \quad = \int_{X_2} e^{pc_1(L_\tau)} (1 - e^{-[Z]}) \text{td}(T_{X_2})$$

$$(2.23) \quad = \frac{c_1(L_\tau)^3 \cdot Z}{3!} p^3 + O(p^2).$$

This ambient computation does not require Z to be smooth or normal.

We next show $H^2(Z, L_\tau^p|_Z) = 0$. Lemma 2.3, the projection formula, and Leray yield

$$H^2(X_2, L_\tau^p) \simeq H^2(\mathbb{P}^2, E_{2,3p}\overline{T}^* \otimes \mathcal{O}(-p\tau)) = 0,$$

where Theorem 2.8 is applied with $\delta = \tau/(3\kappa) < 1/3$.

Using (2.18),

$$L_\tau^p(-Z) \simeq \mathcal{O}_{X_2}(2p - b_1, p - b_2) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(t - p\tau).$$

The tempting attempt to apply the preceding relative vanishing directly to this bundle is incorrect in general: on the fiber $\mathbb{F}_3$ one must control $a_1 - 2a_2$, and here that number equals $2b_2 - b_1$, which may be arbitrarily negative. We instead prove the required H^3 vanishing by a filtration.

Write $q = \pi_{2,1} : X_2 \rightarrow X_1$ and $\pi = \pi_{1,0} : X_1 \rightarrow \mathbb{P}^2$. For $p \gg 1$ one has $p - b_2 \geq 0$, whence

$$R^r q_* L_\tau^p(-Z) = 0 \quad (r \geq 1)$$

and

$$q_* L_\tau^p(-Z) = \mathcal{O}_{X_1}(2p - b_1) \otimes \text{Sym}^{p-b_2} V_1^* \otimes \pi^* \mathcal{O}(t - p\tau).$$

The dual relative tangent sequence

$$(2.24) \quad 0 \longrightarrow \mathcal{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi^* \overline{K} \otimes \mathcal{O}_{X_1}(-2) \longrightarrow 0$$

induces a filtration whose quotients, indexed by $0 \leq j \leq p - b_2$, are

$$\mathcal{O}_{X_1}(3p - m - 3j) \otimes \pi^* (\overline{K}^j \otimes \mathcal{O}(t - p\tau)).$$

Put $\ell = p - j$. The same quotients become

$$(2.25) \quad \mathcal{G}_{\ell,p} = \mathcal{O}_{X_1}(3\ell - m) \otimes \pi^* \mathcal{O}((\kappa - \tau)p + t - \kappa\ell), \quad b_2 \leq \ell \leq p.$$

If $3\ell - m \geq -1$, then $R^1 \pi_* \mathcal{O}_{X_1}(3\ell - m) = 0$, so $H^3(X_1, \mathcal{G}_{\ell,p}) = 0$ by Leray. The remaining indices satisfy

$$b_2 \leq \ell \leq \left\lfloor \frac{m-2}{3} \right\rfloor;$$

this is a fixed finite set, independent of p . For each such ℓ , the projection formula gives

$$R^1 \pi_* \mathcal{G}_{\ell,p} = R^1 \pi_* \mathcal{O}_{X_1}(3\ell - m) \otimes \mathcal{O}((\kappa - \tau)p + t - \kappa\ell).$$

The first factor is fixed, while $\kappa - \tau > 0$ because $\tau < 3 \leq \kappa$. Serre vanishing therefore kills its H^2 for $p \gg 1$. Leray, followed by induction through the filtration (2.24), yields

$$(2.26) \quad H^3(X_2, L_\tau^p(-Z)) = 0.$$

The divisor exact sequence now gives $H^2(Z, L_\tau^p|_Z) = 0$. Combining this with (2.21), we obtain

$$h^0(Z, L_\tau^p|_Z) = \chi(Z, L_\tau^p|_Z) + h^1(Z, L_\tau^p|_Z) + h^3(Z, L_\tau^p|_Z) \geq \chi(Z, L_\tau^p|_Z) \gg p^3,$$

which proves that the $\mathbb{Q}$ -line bundle $L_\tau|_Z$ is big.

Choose p in the same sufficiently divisible sequence and a nonzero section

$$\sigma \in H^0(Z, L_\tau^p|_Z), \quad \tilde{t} = p\tau \in \mathbb{Z}.$$

Since $f^* \omega_1 \equiv 0$ and Lemma 2.4 excludes containment in Γ_2 , one has $f_{[2]}(\mathbb{C}) \subset Z$. Suppose first that the canonical evaluation $s_{\sigma,f}$ is not identically zero. Since

$$L_\tau^p = \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t}),$$

Lemma 2.10 yields

$$\tilde{t} T_f(r) \leq 3p N_f^{[1]}(r, D) + O(\log T_f(r) + \log r) \parallel.$$

The logarithmic derivative lemma makes the final term $o(T_f(r))$ for an algebraically nondegenerate curve. Since $\tilde{t} = p\tau$, this is (2.19).

If instead $s_{\sigma,f} \equiv 0$, the tautological derivative is nonzero away from a discrete set because f is nonconstant. The intrinsic pairing (2.14) therefore implies $f_{[2]}^* \sigma \equiv 0$. Hence $f_{[2]}(\mathbb{C})$ is contained in the proper divisor $\text{div}_Z(\sigma)$ of the irreducible threefold Z . This set has dimension at most two, so its image under $\pi_{2,1} : X_2 \rightarrow X_1$ is a proper algebraic subset of the threefold X_1 . Hence $f_{[1]}$ is algebraically degenerate. $\square$

Remark 2.12. The preceding proof keeps separate the three ingredients required by the argument: the positive cubic intersection, the H^2 and H^3 vanishings needed to obtain bigness, and the intrinsic tautological evaluation and logarithmic-derivative estimate used in (2.19).

2.6. Algebraically degenerate curves. The preceding analytic estimates concern algebraically nondegenerate curves. For the hyperbolicity corollary, the algebraically degenerate case is supplied by the following specialization of Xi Chen's theorem.

For an irreducible curve $\Gamma \not\subset D$, let $\nu : \tilde{\Gamma} \rightarrow \Gamma$ be the normalization and define

$$i(\Gamma, D) := \# \text{Supp } \nu^* D.$$

Theorem 2.13 (Xi Chen). *For a very general ordered pair of smooth transverse plane curves of total degree $s = d_1 + d_2$ and every reduced irreducible curve $\Gamma \not\subset D$,*

$$(2.27) \quad 2g(\tilde{\Gamma}) - 2 + i(\Gamma, D) \geq (s - 4) \deg \Gamma.$$

Source of the specialization. Xi Chen proves for a very general reducible plane divisor $D = \bigcup D_j$ of total degree d that

$$2g(\tilde{\Gamma}) - 2 + i(\Gamma, D) \geq (d - 4) \deg \Gamma.$$

Taking a very general ordered pair of degrees (d_1, d_2) gives $d = s$ and hence (2.27); see [1, Corollary 1.11]. $\square$

Corollary 2.14. *If $s \geq 5$, then for a very general pair there is no nonconstant algebraically degenerate entire curve in $\mathbb{P}^2 \setminus D$.*

Proof. If $f(\mathbb{C})$ is contained in $\Gamma \not\subset D$, it lifts to

$$\tilde{f} : \mathbb{C} \rightarrow \tilde{\Gamma} \setminus \text{Supp } \nu^* D.$$

By (2.27),

$$2g(\tilde{\Gamma}) - 2 + i(\Gamma, D) > 0,$$

so the punctured Riemann surface on the right is hyperbolic. Hence $\tilde{f}$ and f are constant. $\square$

3. LOGARITHMIC CHERN NUMBERS AND THE DEGREE PHASE DIAGRAM

Two ratios organize the proof. The Riemann–Roch threshold ρ_{RR} permits the initial choice $t/m < \rho_{\text{RR}}$; the Demailly–El Goul threshold ρ_{DEG} determines when an irreducible factor lies in the zero-locus range. Keeping both degrees symbolic isolates the six pairs for which the remaining small twists require finite verification.

3.1. The two-component Chern data. For $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ one has

$$\bar{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa), \quad \bar{c}_1 = \kappa h, \quad \kappa = d_1 + d_2 - 3.$$

Using

$$c(\Omega_{\mathbb{P}^2}^1(\log D)) = c(\Omega_{\mathbb{P}^2}^1) \prod_{i=1}^2 (1 + [C_i] + [C_i]^2)$$

and $[C_i] = d_i h$, we obtain

$$(3.1) \quad \bar{c}_1^2 = \kappa^2, \quad \bar{c}_2 = d_1^2 + d_2^2 + d_1 d_2 - 3(d_1 + d_2) + 3, \quad \bar{c}_1 \cdot h = \kappa.$$

Two combinations will recur:

$$(3.2) \quad \bar{c}_1^2 - \bar{c}_2 = d_1 d_2 - 3(d_1 + d_2) + 6,$$

$$(3.3) \quad \begin{aligned} 13\bar{c}_1^2 - 9\bar{c}_2 &= A(d_1, d_2) \\ &= 4(d_1^2 + d_2^2) + 17d_1 d_2 - 51(d_1 + d_2) + 90. \end{aligned}$$

The first is positive in the classical one-jet range $d_1 \geq 5$, or $d_1 = 4, d_2 \geq 7$. The second is positive for every $d_1, d_2 \geq 3$; its minimum there is $A(3, 3) = 9$. On the degree-two boundary,

$$(3.4) \quad A(2, n) = 4n^2 - 17n + 4,$$

which is positive exactly for integers $n \geq 5$. On the degree-one boundary,

$$(3.5) \quad A(1, n) = 4n^2 - 34n + 43,$$

which, for integers $n \geq 2$, is positive exactly when $n \geq 7$.

3.2. Riemann–Roch threshold. This is the standard invariant-jet Riemann–Roch calculation introduced in Demailly’s foundational note [2]. We spell out only the two-component specialization of the leading coefficient because it determines the effective range used later. By (2.2), additivity of the Chern character gives

$$(3.6) \quad \chi(E_{2,m}\bar{T}^* \otimes \bar{K}^{-\delta m}) = \sum_{j=0}^{\lfloor m/3 \rfloor} \chi(S^{m-3j}\bar{T}^* \otimes \bar{K}^{j-\delta m}).$$

Let x_1, x_2 be the formal Chern roots of $\bar{T}^*$. For $q = m - 3j$,

$$\text{ch}(S^q\bar{T}^*) = \sum_{a=0}^q e^{ax_1 + (q-a)x_2}.$$

Keeping terms of cohomological degree two, multiplying by $e^{(j-\delta m)(x_1+x_2)} \text{td}(T_{\mathbb{P}^2})$, and integrating over $\mathbb{P}^2$ computes each summand in (3.6). The Todd terms of positive degree contribute only $O(m^3)$ after summation. For the degree-four term, the elementary identities

$$\sum_{a=0}^q 1 = q + 1, \quad \sum_{a=0}^q a = \frac{q(q+1)}{2}, \quad \sum_{a=0}^q a^2 = \frac{q(q+1)(2q+1)}{6}$$

followed by the sums in $0 \leq j \leq \lfloor m/3 \rfloor$ give

$$(3.7) \quad \sum_{j=0}^{\lfloor m/3 \rfloor} \int_{\mathbb{P}^2} [\text{ch}(S^{m-3j}\bar{T}^*) e^{(j-\delta m)\bar{c}_1}]_2 = \frac{m^4}{648} ((54\delta^2 - 48\delta + 13)\bar{c}_1^2 - 9\bar{c}_2) + O(m^3).$$

Define

$$Q_{\text{RR}}(\delta) = 54\bar{c}_1^2\delta^2 - 48\bar{c}_1^2\delta + (13\bar{c}_1^2 - 9\bar{c}_2).$$

Then, for $t = (\bar{c}_1 \cdot h)\delta m$,

$$(3.8) \quad \chi(\mathbb{P}^2, E_{2,m}\bar{T}^* \otimes \bar{K}^{-\delta m}) = \frac{Q_{\text{RR}}(\delta)}{648} m^4 + O(m^3).$$

Thus the leading coefficient is positive for sufficiently small positive δ . In the ratio $\rho = t/m = \kappa\delta$, its smaller positive root is

$$(3.9) \quad \rho_{\text{RR}}(d_1, d_2) = \frac{\kappa}{18} \left(8 - \sqrt{54\frac{\bar{c}_2}{\bar{c}_1^2} - 14} \right) > 0.$$

The strict positivity is equivalent to $A(d_1, d_2) > 0$. For two cubics this specializes to $(8 - \sqrt{58})/6 > 1/16$.

3.3. Zero-locus threshold. Set $\rho = t/m$. Using the projective-bundle relations on the logarithmic Demailly–Sample tower, the cubic intersection occurring in Theorem 2.11 is

$$(3.10) \quad (2u_1 + u_2 - \tau h)^3 \cdot Z = mQ_{\text{DEG}}(\rho, \tau),$$

where, as in (2.12),

$$Q_{\text{DEG}}(\rho, \tau) = 3\tau^2 + (9\rho - 12(\bar{c}_1 \cdot h))\tau + (13\bar{c}_1^2 - 9\bar{c}_2) - 12(\bar{c}_1 \cdot h)\rho.$$

Here is the intersection calculation. Put $\Theta = 2u_1 + u_2$ and write $[Z] = b_1u_1 + b_2u_2 - \tau h$, where $b_1 + b_2 = m$. Apply successively the rank-two projective-bundle formula

$$\pi_*(u) = 1, \quad \pi_*(u^2) = -c_1(V), \quad \pi_*(u^3) = c_1(V)^2 - c_2(V)$$

first to $\pi_{2,1}$ and then to $\pi_{1,0}$. With our convention that $\mathbb{P}(V)$ parametrizes lines, the intermediate identities are

$$u_1^2 - \bar{c}_1u_1 + \bar{c}_2 = 0, \quad c_1(V_1) = u_1 - \bar{c}_1, \quad c_2(V_1) = 2\bar{c}_2 - \bar{c}_1u_1.$$

Expanding with these relations gives the following three identities on the base:

$$(3.11) \quad \Theta h^2 \cdot Z = m,$$

$$(3.12) \quad \Theta^2 h \cdot Z = 4m(\bar{c}_1 \cdot h) - 3t,$$

$$(3.13) \quad \Theta^3 \cdot Z = m(13\bar{c}_1^2 - 9\bar{c}_2) - 12t(\bar{c}_1 \cdot h).$$

The terms involving b_1 and b_2 combine through $b_1 + b_2 = m$; this is why the answer is independent of the chosen irreducible factor. Since $h^3 = 0$ on the surface base, expansion of $(\Theta - \tau h)^3 \cdot Z$ and substitution of (3.11)–(3.13) yield

$$\begin{aligned} (\Theta - \tau h)^3 \cdot Z &= \Theta^3 \cdot Z - 3\tau\Theta^2 h \cdot Z + 3\tau^2\Theta h^2 \cdot Z \\ &= mQ_{\text{DEG}}(\rho, \tau), \end{aligned}$$

which proves (3.10) directly. At $\tau = 0$ this is positive exactly when

$$(3.14) \quad \rho < \rho_{\text{DEG}} = \frac{13\bar{c}_1^2 - 9\bar{c}_2}{12(\bar{c}_1 \cdot h)}.$$

Using (3.3),

$$(3.15) \quad \rho_{\text{DEG}}(d_1, d_2) = \frac{A(d_1, d_2)}{12\kappa}.$$

Solving this quadratic recovers the smaller positive root $\tau_1(\rho)$ in (2.13). For (3, 3) this is

$$(3.16) \quad \tau_1(\rho) = 6 - \frac{3}{2}\rho - \frac{1}{2}\sqrt{132 - 24\rho + 9\rho^2}.$$

Equation (3.10) supplies the positive leading term; bigness and the second differential follow from the cohomological theorem, not from this number alone.

3.4. Why the proved range has six finite boundary pairs. For a genuine second-level component one has $b_2 > 0$ and $b_1 \geq 2b_2$, hence $m = b_1 + b_2 \geq 3$. If $t \leq 3$, then $t/m \leq 1$. Consequently no finite vanishing is needed whenever $\rho_{\text{DEG}} > 1$. We apply this observation only after the Riemann–Roch positivity condition $A(d_1, d_2) > 0$ has supplied the initial differential. Direct calculation gives

$$\rho_{\text{DEG}}(3, 3) = \frac{1}{4}, \quad \rho_{\text{DEG}}(3, 4) = \frac{37}{48}, \quad \rho_{\text{DEG}}(3, 5) = \frac{73}{60} > 1,$$

and $\rho_{\text{DEG}} > 1$ for every other pair with $3 \leq d_1 \leq d_2$, except (3, 3) and (3, 4). On the degree-two boundary,

$$\rho_{\text{DEG}}(2, 5) = \frac{19}{48}, \quad \rho_{\text{DEG}}(2, 6) = \frac{23}{30},$$

whereas $\rho_{\text{DEG}}(2, n) > 1$ for $n \geq 7$. This proves the phase diagram on the degree-two boundary. Finally,

$$\rho_{\text{DEG}}(1, 8) = \frac{3}{8}, \quad \rho_{\text{DEG}}(1, 9) = \frac{61}{84}, \quad \rho_{\text{DEG}}(1, 10) = \frac{103}{96} > 1,$$

and $\rho_{\text{DEG}}(1, n) > 1$ for every $n \geq 10$. Together these calculations reduce the asserted line-component range $n \geq 8$ to (1, 8) and (1, 9). The omitted formal endpoint (1, 7) has $A(1, 7) = 1$ and $\rho_{\text{DEG}}(1, 7) = 1/60$; it is not a seventh certified boundary pair, because its required vanishing profile is neither proved nor claimed here. Remark 1.1 records that separate problem. Thus, within the theorem’s stated range, the pairs with $A(d_1, d_2) > 0$ but $\rho_{\text{DEG}} \leq 1$ are exactly (3, 3), (3, 4), (2, 5), (2, 6), (1, 8), (1, 9). These are precisely the six finite lists in Lemma 1.2.

4. THE FIRST NEGATIVELY TWISTED DIFFERENTIAL

The construction has three steps. Riemann–Roch and H^2 vanishing first give one section with a degree pair independent of the parameter. The Demailly–El Goul reduction then selects a prime horizontal component with no loss in the twist-to-weight ratio. Finally, cohomology and base change spread its canonical reduced equation over one Zariski-open subset of the ordered parameter space. The output is therefore a fixed numerical type (m, t, b_1, b_2) and one algebraic family of prime horizontal divisors; these are exactly the uniform data used by the mixed-field construction.

Proposition 4.1 (Uniform existence). *Fix a rational number*

$$(4.1) \quad 0 < \rho_0 < \rho_{\text{RR}}(d_1, d_2).$$

There exist fixed positive integers M, T with $T/M = \rho_0$ such that for every $a \in \mathcal{S}_{d_1, d_2}^{\text{snc}}$ one has

$$(4.2) \quad H^0(\mathbb{P}^2, E_{2, M} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-T)) \neq 0.$$

Proof. Put $\delta_0 = \rho_0/(\bar{c}_1 \cdot h) = \rho_0/\kappa$. Write $\rho_0 = T_0/M_0$ with positive integers M_0, T_0 . Choose a sufficiently large N such that NT_0/κ is integral, and set

$$M = NM_0, \quad T = NT_0.$$

Then $\delta_0 M = T/\kappa \in \mathbb{Z}$, so the twist $\bar{K}^{-\delta_0 M}$ is an honest line bundle. All smooth transverse pairs of degrees (d_1, d_2) have the same logarithmic Chern numbers (3.1). Hence the Riemann–Roch polynomial in (3.8), including its lower-order coefficients, is independent of the parameter $a \in \mathcal{S}_{d_1, d_2}^{\text{snc}}$. Since $Q_{\text{RR}}(\delta_0) > 0$, one may choose N once and for all so that this Euler characteristic is positive for every such a . Theorem 2.8, applied with the same δ_0 , also holds for every D_a . Therefore

$$h^0 = h^1 + \chi - h^2 \geq \chi > 0$$

for every $a \in \mathcal{S}_{d_1, d_2}^{\text{snc}}$, proving (4.2). $\square$

The following irreducible-component reduction is due to Demailly–El Goul [3, Lemma 3.3]; we use the implementation in [5, Section 4]. The reduction itself is unchanged in the two-component setting. We reproduce the short argument only to fix the notation and the ratio inequality needed below.

Lemma 4.2 (Irreducible-component reduction). *From a nonzero section of weighted degree M and twist T one may choose an irreducible horizontal factor*

$$0 \neq \omega_1 \in H^0(\mathbb{P}^2, E_{2, m} \bar{T}^* \otimes \mathcal{O}(-t))$$

whose zero divisor on X_2 has the form $Z + b\Gamma_2$, with Z irreducible, reduced, and horizontal, and such that

$$\frac{t}{m} \geq \frac{T}{M}.$$

Writing the horizontal component as

$$Z \sim b_1 u_1 + b_2 u_2 - th, \quad b_1 \geq 2b_2 \geq 0, \quad b_1 + b_2 = m,$$

one has the following alternative: if $b_2 = 0$, then Z is pulled back from a divisor on X_1 ; if $b_2 > 0$, it is a genuine second-level horizontal divisor. In particular, applying the lemma to the fixed pair (M, T) supplied by Proposition 4.1 yields an irreducible ω_1 satisfying

$$\frac{t}{m} \geq \rho_0, \quad \text{and hence} \quad \frac{m}{t} \leq \rho_0^{-1}.$$

Proof. Via (2.3), the original section is represented on X_2 by an effective divisor

$$Z_\omega \sim Mu_2 - Th.$$

Write its irreducible decomposition as

$$Z_\omega = \sum_j p_j Z_j, \quad Z_j \sim b_{1,j}u_1 + b_{2,j}u_2 - t_j h.$$

The standard classification of effective divisors on the second Demailly–Semple tower gives three possibilities:

- (i) $(b_{1,j}, b_{2,j}) = (0, 0)$ and Z_j is pulled back from the base; then $t_j \leq 0$;
- (ii) $(b_{1,j}, b_{2,j}) = (-1, 1)$, $t_j = 0$, and $Z_j = \Gamma_2$;
- (iii) Z_j is horizontal and $b_{1,j} \geq 2b_{2,j} \geq 0$, with $b_{1,j} + b_{2,j} > 0$.

For a horizontal component put

$$m_j = b_{1,j} + b_{2,j} > 0.$$

Since the class of Z_ω has $u_1 + u_2$ -weight M , while the first two types have weight zero, comparison of divisor classes gives

$$\sum_{j \text{ horizontal}} p_j m_j = M.$$

Comparison of the base twists gives

$$\sum_j p_j t_j = T.$$

The base-pullback components have $t_j \leq 0$ and Γ_2 has $t_j = 0$, so

$$\sum_{j \text{ horizontal}} p_j t_j \geq T.$$

Consequently at least one horizontal component satisfies

$$\frac{t_j}{m_j} \geq \frac{T}{M}.$$

Fix such a component and rename it Z . Because $\mathcal{O}_{X_2}(\Gamma_2) = \mathcal{O}_{X_2}(-1, 1)$, one has

$$Z + b_{1,j}\Gamma_2 \sim (b_{1,j} + b_{2,j})u_2 - t_j h = m_j u_2 - t_j h.$$

Thus, if s_j is the canonical reduced equation of the prime Cartier divisor Z and γ is the canonical section of Γ_2 , then $\gamma^{b_{1,j}} s_j$ is a section of $\mathcal{O}_{X_2}(m_j) \otimes \pi_{2,0}^* \mathcal{O}(-t_j)$, which corresponds by (2.3) to an invariant jet differential

$$\omega_j \in H^0\left(\mathbb{P}^2, E_{2,m_j} \bar{T}^* \otimes \mathcal{O}(-t_j)\right).$$

After saturation by the fixed vertical factor $\gamma^{b_{1,j}}$, its horizontal zero divisor is exactly the reduced prime Z ; no extraction of a root of a global section is involved. When $b_{2,j} = 0$, the classification identifies Z_j with the inverse image under $\pi_{2,1}$ of a divisor on X_1 . The ratio inequality proves the final assertion. $\square$

The following family-extension step follows the algebraic spread-out construction in [5, Section 4], applied to the ordered two-component parameter space. The only change is the base: the generic freeness, cohomology-and-base-change, and geometric-integrality arguments are the same.

Proposition 4.3 (Algebraic extension of the chosen factor). *Let (M, T) be the fixed integers supplied by Proposition 4.1. Then there exist integers (m, t) with $t/m \geq T/M$, integers*

$$b_1 \geq 2b_2 \geq 0, \quad b_1 + b_2 = m,$$

and a nonempty Zariski-open subset $\mathcal{U} \subset \mathcal{S}_{d_1, d_2}^{\text{snc}}$ with the following property. Let

$$p_{\mathcal{U}} : \mathcal{X}_2 \longrightarrow \mathcal{U}, \quad \Pi : \mathcal{X}_2 \longrightarrow \mathbb{P}^2 \times \mathcal{U}$$

be the relative second logarithmic Semple tower and its projection to level zero. There is a regular algebraic section

$$f \in H^0(\mathcal{X}_2, \mathcal{L}), \quad \mathcal{L} = \mathcal{O}_{\mathcal{X}_2}(b_1, b_2) \otimes \Pi^* \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

whose zero scheme $\mathcal{Z} = (f = 0)$ is flat over $\mathcal{U}$ and has geometrically integral fibers. If γ denotes the canonical section of the relative vertical divisor Γ_2 , then

$$\omega_{1,a} = \gamma_a^{b_1} s_a \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)), \quad a \in \mathcal{U},$$

is an algebraic family of invariant jet differentials, and its horizontal zero divisor is the irreducible reduced divisor $Z_a = \mathcal{Z}|_{X_{2,a}}$.

Proof. Work first over $S = \mathcal{S}_{d_1, d_2}^{\mathrm{snc}}$, and denote its relative second logarithmic Semple tower by $p_S : \mathcal{X}_{2,S} \rightarrow S$. For every triple (b_1, b_2, t) consider the algebraic line bundle

$$\mathcal{L}_{b_1, b_2, t} = \mathcal{O}_{\mathcal{X}_{2,S}}(b_1, b_2) \otimes \Pi_S^* \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t).$$

The sheaf $(p_S)_* \mathcal{L}_{b_1, b_2, t}$ is coherent. Generic freeness and generic cohomology and base change give a dense Zariski-open set $S_{b_1, b_2, t} \subset S$ on which it is locally free and the natural map

$$(4.3) \quad ((p_S)_* \mathcal{L}_{b_1, b_2, t}) \otimes k(a) \longrightarrow H^0(X_{2,a}, \mathcal{L}_{b_1, b_2, t}|_{X_{2,a}})$$

is an isomorphism. There are only countably many triples. Since the base field is the uncountable field $\mathbb{C}$, an irreducible variety of finite type over $\mathbb{C}$ cannot be the union of countably many proper closed subsets. Equivalently, every countable intersection of dense Zariski-open subsets has a complex point. We may therefore choose

$$a_0 \in \bigcap_{b_1, b_2, t} S_{b_1, b_2, t}.$$

This countable intersection is used only to select one numerical type and one prime factor; after that choice, the construction is spread over a single Zariski-open neighborhood. By Proposition 4.1, this fiber carries a section of the fixed weighted degree M and fixed twist T . Apply Lemma 4.2 on the fiber over a_0 and choose its reduced horizontal equation

$$0 \neq s_{a_0} \in H^0(X_{2,a_0}, \mathcal{L}_{b_1, b_2, t}|_{X_{2,a_0}})$$

with $b_1 \geq 2b_2 \geq 0$, $b_1 + b_2 = m$, and $t/m \geq T/M$. For this particular triple, the base-change isomorphism above identifies s_{a_0} with a vector in the fiber of the locally free sheaf $(p_S)_* \mathcal{L}_{b_1, b_2, t}$. After shrinking to an affine trivializing neighbourhood, extend that vector to a regular local section of the pushforward. The adjunction evaluation map

$$p_S^*(p_S)_* \mathcal{L}_{b_1, b_2, t} \longrightarrow \mathcal{L}_{b_1, b_2, t}$$

then produces a regular section f on the relative tower after replacing S by an affine Zariski-open neighborhood of a_0 .

The section s_{a_0} is nonzero and its zero scheme is the selected irreducible reduced horizontal divisor. Thus it is an integral Cartier divisor; over the algebraically closed field $\mathbb{C}$ it is also geometrically integral. Nonvanishing on every irreducible component of a fiber is an open condition, so after shrinking, $\mathcal{Z} = (f = 0)$ is a relative effective Cartier divisor. It is then flat over S because both $\mathcal{X}_{2,S}$ and the invertible ideal $\mathcal{L}^{-1}$ are flat and

$$0 \longrightarrow \mathcal{L}^{-1} \xrightarrow{f} \mathcal{O}_{\mathcal{X}_{2,S}} \longrightarrow \mathcal{O}_{\mathcal{Z}} \longrightarrow 0$$

remains exact on the fibers. Finally, geometric integrality of fibers is open for a proper flat morphism of finite presentation. A second shrinking therefore makes every Z_a geometrically integral. The relative divisor Γ_2 and its canonical section γ are algebraic. Consequently $\gamma^{b_1} f$ is a regular relative section of

$$\mathcal{O}_{\mathcal{X}_2}(m) \otimes \Pi^* \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

and the relative direct-image formula gives the asserted algebraic family $\omega_{1,a}$. This proves all statements after naming the resulting open set $\mathcal{U}$. $\square$

From now on we fix the irreducible differential supplied by Lemma 4.2 and denote its weighted degree and twist by (m, t) .

5. MIXED $\mathcal{O}_{\mathbb{P}^2}(3)$ SLANTED VECTOR FIELDS

The purpose of this section is precise. Starting with the relative prime divisor selected in Proposition 4.3, we shall construct an honest global invariant slanted field of base twist $\mathcal{O}_{\mathbb{P}^2}(3)$ whose natural action on the equivariant jet-polynomial equation cuts that divisor properly. Restriction to a parameter fibre then gives the second global jet differential of Proposition 5.7. The construction is lengthy because the local algebra, projective descent, and invariant action must all be compatible.

We retain the standing notation

$$\mathcal{Z} = (f = 0), \quad \omega_{1,a} = \gamma_a^{b_1} s_a$$

from Proposition 4.3. Thus f is the Γ_2 -saturated horizontal equation, with the fixed vertical factor removed. It is f , not the product $\gamma^{b_1} f$, on which the invariant field first acts; the factor $\gamma_a^{b_1}$ is reattached after restriction in Phase VI.

The compact template. The compact pole-three argument is easiest to remember on the invariant fiber ring

$$K[u, v, W], \quad \text{wt}(u) = \text{wt}(v) = 1, \quad \text{wt}(W) = 3.$$

If every available slanted derivative vanished normally along a prime divisor ($P = 0$), then the principal ideal (P) would be stable under the corresponding residual derivations. In the compact case an acceleration field first supplies a nonzero multiple of ∂_W , so that P cannot depend on W ; a matrix field then acts on the remaining binary form and contradicts irreducibility in positive weight. This two-step contradiction is the expository model for what follows.

The two-component logarithmic adaptation. The compact fields cannot simply be copied. Each logarithmic component contributes its own root equation and its own value, gradient, and Hessian conditions. The same matrix motion T and the same quadratic acceleration C must therefore solve two Hessian-cancellation equations at once. On the transverse open set this determines a common correction $C = C_T$ through the inverse of the two-row gradient matrix; the resulting residual derivations are the fields D_T in (5.21). Unlike the compact model, no independent acceleration shear is inserted at this point. Instead, the commutators of D_H, D_E, D_F produce a nonzero polynomial multiple of ∂_W in (5.43). That curvature direction removes the W -dependence, after which the full binary $\mathfrak{sl}_2$ -action forces P to be constant. This is the essential two-component adaptation.

The ingredients have distinct sources. Finite-difference coefficient fields go back to [13, Section 3]; the compact matrix lift appears in [15, Lemma 1.1], and its logarithmic form in [16, Lemma 5]. The quadratic-acceleration field, its common-twist combination, and the resulting independence argument are adapted from an announced revision of Cui–Hou–Liu–Xie [8]. The cited public version does not yet contain that pole-order-three package, so we give the complete argument used here. The new work is the coupling of two component equations while retaining one pole-three twist: this requires two root gauges, simultaneous Hessian cancellation, regularity at the logarithmic corner, and separate generators when a component has degree at most two.

For navigation, the proof is divided into six phases.

- (1) Identify the invariant fiber ring $K_0[u, v, W]$ from the two logarithmic jet equations.
- (2) Use third differences to absorb precisely the coefficient variations whose Taylor expansion starts in order three.
- (3) Couple the matrix and acceleration data, solve the two Hessian equations simultaneously, and identify the residual derivations D_T .

- (4) Descend the honest polynomial fields through both root gauges and all Semple charts with the common base twist $\mathcal{O}_{\mathbb{P}^2}(3)$.
- (5) Use commutator curvature and the binary $\mathfrak{sl}_2$ -action to rule out a positive-weight invariant prime.
- (6) Choose a field whose global derivative cuts the prime divisor properly and restrict it to obtain the second global jet differential.

The distinction between an *honest field* and a *residual derivation* is important throughout: the former is global and polynomial, whereas the latter is only a generic-field linear combination used to test invariance of the prime ideal. The number three also has three related but distinct meanings below. We use third differences because a tangent correction must have zero Taylor coefficients through order two; projective overlap then gives pole order at most three; homogenization records this bound as the twist $\mathcal{O}_{\mathbb{P}^2}(3)$. The construction still concerns invariant two-jets, not three-jets.

5.1. Phase I: two roots and the invariant fiber ring. On the affine chart $Z_0 \neq 0$, write the two universal equations as

$$f_i(z, a_i) = \sum_{|\alpha| \leq d_i} a_{i,\alpha} z^\alpha, \quad z = (z_1, z_2), \quad i = 1, 2.$$

Near a crossing of the two components one must keep the two logarithmic normal directions separate. We therefore introduce two roots

$$(5.1) \quad w_i^{d_i} + f_i(z, a_i) = 0, \quad i = 1, 2.$$

This construction is made first on affine charts of the two coefficient cones. Equivalently, over such a chart it is the fiber product of the two cyclic covers associated with the local trivializations of $\mathcal{O}_{\mathbb{P}^2}(d_i)$. The chartwise covers are auxiliary: deck equivariance, the product coefficient–root gauge in Lemma 5.2, and reparametrization equivariance below are precisely the descent data that produce vector fields on the relative logarithmic Semple tower. Thus no choice of a global root of either defining equation enters the theorem. The cover is smooth over the simple-normal-crossing locus, including above $C_1 \cap C_2$. Its deck group is $\mu_{d_1} \times \mu_{d_2}$; when $d_i = 1$ the corresponding deck factor is trivial and the equation $w_i + f_i = 0$ simply defines a graph. The coefficient–root radial gauge is nevertheless still present. The two independent radial fields on the coefficient cones are

$$(5.2) \quad R_i = d_i \sum_{|\alpha| \leq d_i} a_{i,\alpha} \partial_{a_{i,\alpha}} + w_i \partial_{w_i}.$$

Write the ordinary first and second derivatives as

$$\xi = (u, v), \quad \eta = (U, V),$$

and write the logarithmic root derivatives as

$$w'_i = w_i \chi_i, \quad w''_i = w_i(\psi_i + \chi_i^2).$$

Differentiating (5.1) zero, one, and two times gives

$$(5.3) \quad E_{i0} = w_i^{d_i} + f_i,$$

$$(5.4) \quad E_{i1} = d_i w_i^{d_i} \chi_i + df_i(\xi),$$

$$(5.5) \quad E_{i2} = d_i w_i^{d_i} \psi_i + d_i^2 w_i^{d_i} \chi_i^2 + df_i(\eta) + \text{Hess}(f_i)(\xi, \xi).$$

Here is the tangency test used throughout the construction. If a field has only an i -th coefficient component, represented by the polynomial $Q_i(Y)$, then its action on the three equations is

$$\begin{aligned} \delta E_{i0} &= Q_i(z), \\ \delta E_{i1} &= dQ_i|_z(\xi), \\ \delta E_{i2} &= dQ_i|_z(\eta) + \text{Hess}(Q_i)|_z(\xi, \xi). \end{aligned}$$

Thus a coefficient correction with $j_z^2 Q_i = 0$ is tangent to the entire two-jet system, independently of the chosen jet. This single observation is the reason that third-order Taylor remainders can be used below without repeating the tangency calculation for every coefficient field. On the dense open set $f_1 f_2 \neq 0$, these equations eliminate χ_i, ψ_i . The unipotent part of the two-jet reparametrization group acts by

$$(u, v, U, V) \mapsto (u, v, U + \lambda u, V + \lambda v).$$

Putting $W = uV - vU$, one obtains

$$(5.6) \quad \mathbb{C}[u, v, U, V]^{G_{2, \text{unip}}} = \mathbb{C}[u, v, W], \quad \text{wt}(u) = \text{wt}(v) = 1, \quad \text{wt}(W) = 3.$$

Indeed, after localizing at u , the infinitesimal generator becomes $u\partial_U$ in the coordinates (u, v, U, W) . Its kernel is $\mathbb{C}[u, v, W]_u$, and the absence of a negative power of u follows by reducing a putative numerator modulo u . This also proves (5.6) over every characteristic-zero extension field.

We now specify the generic field used below. On fixed affine coefficient charts, let

$$\mathcal{B}^\circ = \left\{ (z, a_1, a_2) : f_1 f_2 \det \begin{pmatrix} \partial_1 f_1 & \partial_2 f_1 \\ \partial_1 f_2 & \partial_2 f_2 \end{pmatrix} \neq 0 \right\}.$$

It is a nonempty open subset of an affine space: for instance $f_1 = 1 + z_1$ and $f_2 = 1 + z_2$ at $z = 0$ give a point of it. Hence it is irreducible. We set

$$(5.7) \quad K_0 := \mathbb{C}(\mathcal{B}^\circ).$$

The affine two-jet incidence over it is cut out by the six equations $E_{i\nu} = 0$ in the variables $(w_i, \chi_i, \psi_i, u, v, U, V)$. On $f_1 f_2 \neq 0$, equations E_{i0} give a finite root extension and E_{i1}, E_{i2} eliminate χ_i, ψ_i ; its function field is therefore a finite extension of $K_0(u, v, U, V)$. Taking invariants of the unipotent reparametrization leaves $K_0[u, v, W]$ after descent from that finite extension. Every denominator in the generic residual coefficients divides a product of

$$(5.8) \quad f_1 f_2, \quad \det(g_1, g_2), \quad \ell_1(a_1) \ell_2(a_2),$$

on the corresponding affine incidence chart. The projective and Semple transition coordinates are different: their zero sets are not removed by a parameter shrinking. They are covered by the overlap and boundary calculations in Phase IV.

Thus Phase I produces the arena for the contradiction: an explicitly defined generic field K_0 , the invariant ring $K_0[u, v, W]$, and a recorded list of the genuine generic-incidence denominators. The next two phases construct honest fields whose generic residual actions operate on this ring.

5.2. Phase II: third differences and one-stage Taylor reduction. We now build, separately in the two coefficient blocks, the third-difference Taylor package. Its finite-difference antecedent is [13, Section 3]; the exact third-order organization is adapted from the forthcoming revision of [8]. The point that requires care here is that the two corrections will subsequently be coupled by the common base and jet-variable motion.

For $d_i \geq 3$ and multi-indices $\alpha \geq \gamma$ with $|\alpha| \leq d_i$ and $|\gamma| = 3$, define the third-difference field in the i -th coefficient block by

$$(5.9) \quad \mathcal{D}_{i, \alpha, \gamma} = \sum_{\beta \leq \gamma} (-1)^{|\beta|} \binom{\gamma}{\beta} z^\beta \partial_{a_{i, \alpha - \beta}}.$$

Its coefficient polynomial is $Y^{\alpha - \gamma}(Y - z)^\gamma$. It belongs to the third power $\mathfrak{m}_z^3$ of the maximal ideal at z and hence annihilates all three equations (5.3)–(5.5). Over any characteristic-zero field, the values of these fields span

$$(5.10) \quad \mathfrak{m}_z^3 \cap K[Y_1, Y_2]_{\leq d_i}.$$

This is just the Taylor basis: choose $\gamma \leq \delta$ with $|\gamma| = 3$ and expand $(Y - z)^{\delta - \gamma}$ in powers of Y .

This package is a correction engine, not a complete tangent frame. It can absorb exactly the coefficient polynomials in $\mathfrak{m}_z^3 \cap K[Y_1, Y_2]_{\leq d_i}$, while it contributes no base or jet-variable motion. The missing motions are supplied in Phase III by T , C , and N_i ; their remaining coefficient components will then be returned to this Taylor span.

We shall also use the following elementary reduction for $d_i \geq 2$. Suppose that a polynomial $Q^0(Y; z)$ has Y -degree at most $d_i + 1$, its top homogeneous part is independent of z , and its remaining coefficients have z -degree at most two. For every degree- $(d_i + 1)$ monomial choose once and for all $\gamma \leq \alpha$ with $|\gamma| = 3$, and subtract the corresponding multiple of

$$(Y - z)^\gamma Y^{\alpha - \gamma}.$$

The resulting polynomial Q has Y -degree at most d_i , $Q - Q^0 \in \mathfrak{m}_z^3$, and coefficient functions of z -degree at most three. In particular, Q and Q^0 have the same value, gradient, and Hessian at $Y = z$. For $d_i = 2$, the same reduction lowers degree three to degree two. More generally,

$$\mathfrak{m}_z^3 \cap K[Y_1, Y_2]_{\leq d_i} = 0 \quad (d_i \leq 2),$$

so no third-difference field exists or is needed in degrees one and two. An arbitrary raw polynomial (5.12) need not have degree at most one when $d_i = 1$; the compatible degree-one generators are constructed in Lemma 5.1 below. Thus this stage supplies the degree-compatible coefficient fields; the next stage couples their values through one common base and jet-variable motion.

5.3. Phase III: simultaneous Hessian cancellation and mixed fields. The following construction couples the classical matrix-lift input [15, Lemma 1.1] and [16, Lemma 5] with the quadratic-acceleration package of the forthcoming revision of [8]. In contrast with the one-equation construction, the acceleration tensor below is determined by two simultaneous Hessian equations.

Let $T \in \mathfrak{sl}_2(\mathbb{C})$, let $C : \text{Sym}^2 \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be symmetric, and let $N_i \in (\mathbb{C}^2)^*$. Prescribe

$$(5.11) \quad \delta\xi = T\xi, \quad \delta\eta = T\eta + C(\xi, \xi), \quad \delta\chi_i = N_i(\xi), \quad \delta\psi_i = N_i(\eta).$$

In the i -th coefficient block begin with

$$(5.12) \quad Q_i^0(Y) = -df_i(Y)T(Y - z) + d_i f_i(Y)N_i(Y - z) - \frac{1}{2}df_i(Y)C(Y - z, Y - z).$$

For $d_i \geq 2$, apply the preceding one-stage reduction and denote the degree-at-most- d_i polynomial by Q_i . When $d_i = 1$, use instead the compatible degree-one fields of Lemma 5.1. The vector fields obtained in either way, consisting of (5.11) and the coefficient vector represented by Q_i , satisfy the exact ideal identities

$$(5.13) \quad V(E_{i0}) = 0,$$

$$(5.14) \quad V(E_{i1}) = d_i N_i(\xi)E_{i0},$$

$$(5.15) \quad V(E_{i2}) = d_i N_i(\eta)E_{i0} + 2d_i N_i(\xi)E_{i1}.$$

The identities follow by direct substitution. The acceleration term in (5.12) has zero value and gradient at z and Hessian $-df_i \circ C$; this cancels the variation of $df_i(\eta)$ caused by $C(\xi, \xi)$. Any reduction term lies in $\mathfrak{m}_z^3$ and changes none of these identities; the degree-one identities are checked directly in Lemma 5.1.

We may now verify reparametrization equivariance without a forward reference. For (λ, μ) with $\lambda \neq 0$, the two-jet group acts by

$$\tilde{\xi} = \lambda\xi, \quad \tilde{\eta} = \lambda^2\eta + \mu\xi, \quad \tilde{\chi}_i = \lambda\chi_i, \quad \tilde{\psi}_i = \lambda^2\psi_i + \mu\chi_i.$$

Substitution in (5.11) gives

$$\begin{aligned} \delta\tilde{\xi} &= T\tilde{\xi}, & \delta\tilde{\eta} &= T\tilde{\eta} + C(\tilde{\xi}, \tilde{\xi}), \\ \delta\tilde{\chi}_i &= N_i(\tilde{\xi}), & \delta\tilde{\psi}_i &= N_i(\tilde{\eta}). \end{aligned}$$

Hence the mixed fields are equivariant for both the scaling and unipotent parts of the group and descend to invariant two-jets.

Put

$$F_i = f_i(z), \quad p_i = df_i|_z, \quad g_i = \frac{p_i}{F_i}, \quad K_i = \text{Hess}(\log f_i)|_z = \frac{\text{Hess}(f_i)|_z}{F_i} - \frac{p_i^\top p_i}{F_i^2}.$$

The value of Q_i^0 at z is zero. Its gradient vanishes precisely for

$$(5.16) \quad N_i = \frac{p_i T}{d_i F_i},$$

and, after this choice, its Hessian vanishes precisely when

$$(5.17) \quad g_i \circ C + T^\top K_i + K_i T = 0.$$

This identity is the reason that the matrix and acceleration parts must be treated together.

Work over the field K_0 in (5.7); by construction $g_1 \wedge g_2 \neq 0$. The map

$$G : K_0^2 \longrightarrow K_0^2, \quad G(\tau) = (g_1(\tau), g_2(\tau))$$

is then an isomorphism. For every $T \in \mathfrak{sl}_2(K_0)$, the two equations (5.17) have the unique simultaneous solution

$$(5.18) \quad C_T(\zeta, \zeta) = -G^{-1} \left(\zeta^\top (T^\top K_1 + K_1 T) \zeta \right).$$

Formula (5.18) is the precise replacement for the free compact acceleration. A single component imposes one scalar quadratic identity; the two independent logarithmic gradients assemble the two identities into the invertible map G and thereby determine the vector-valued quadratic correction C_T uniquely.

Lemma 5.1 (Compatible generators for a linear component). *Assume $d_1 = 1$ and $d_2 \geq 2$. Put*

$$x = Y - z, \quad f_1(Y) = F + p(x), \quad p = (p_1, p_2) = df_1, \quad v_p = (-p_2, p_1)^\top.$$

The line block of (5.12) is represented by a polynomial of Y -degree at most one if and only if

$$(5.19) \quad p(C(x, x)) = 2p(x)N_1(x),$$

and then

$$(5.20) \quad Q_1(Y) = (-pT + FN_1)x.$$

There is a finite polynomial family of ten honest fields with this property, split as $3 + 2 + 3 + 2$: three T -fields; two line-normal fields, for which

$$C_N(\zeta, \eta) = N(\zeta)\eta + N(\eta)\zeta, \quad N_1 = N;$$

three kernel-acceleration fields, for which

$$C_{p,q}(\zeta, \zeta) = v_p q(\zeta, \zeta), \quad q \in \{\zeta_1^2, \zeta_1 \zeta_2, \zeta_2^2\};$$

and two second-normal fields, for which only N_2 is nonzero. After the degree- d_2 Taylor reduction in the second block, these ten fields satisfy (5.13)–(5.15). Their span over K_0 contains the residual field determined by (5.16) and (5.18); for that residual field one has $Q_1 = 0$.

Proof. Since f_1 is linear, substitution in (5.12) gives

$$Q_1^0(Y) = -pTx + (F + px)N_1x - \frac{1}{2}p(C(x, x)).$$

Its quadratic part vanishes exactly under (5.19), and the remaining linear polynomial is (5.20). For the line-normal fields, $C_N(x, x) = 2N(x)x$, so the compatibility is automatic and $Q_1 = FN(x)$. For a kernel-acceleration field one has $p(v_p) = 0$, hence $Q_1 = 0$. The T -fields give $Q_1 = -pTx$, while a second-normal field contributes nothing in the line block. Thus all ten fields are represented by legitimate line-coefficient variations, and direct substitution gives

the three tangent-ideal identities. The second block is handled by the degree- d_2 reduction already proved above. Although v_p depends linearly on the coefficients of the line, it is independent of z ; hence the degree and Taylor-remainder bounds used in that reduction are unchanged.

It remains to check that these polynomial fields span the required generic one. For a line,

$$K_1 = -\frac{p^\top p}{F^2}, \quad N_1 = \frac{pT}{F}.$$

Evaluating the first equation in (5.17) on (ζ, ζ) therefore gives

$$p(C_T(\zeta, \zeta)) = 2p(\zeta)N_1(\zeta) = p(C_{N_1}(\zeta, \zeta)).$$

On the generic incidence open $p \neq 0$, and v_p spans $\ker p$. Hence there is a scalar quadratic form q_T over K_0 such that

$$(C_T - C_{N_1})(\zeta, \zeta) = v_p q_T(\zeta, \zeta).$$

These four families therefore span the data (T, C_T, N_1, N_2) . Finally $-pT + FN_1 = 0$, so (5.20) gives $Q_1 = 0$. $\square$

Degree ledger. At the residual choice, $Q_i \in \mathfrak{m}_z^3$ in both blocks. If $d_i \geq 3$, the Taylor span (5.10) cancels Q_i by third differences. If $d_i = 2$, the reduced polynomial lies in $\mathfrak{m}_z^3$ and is therefore already zero. If $d_1 = 1$, the compatible family of Lemma 5.1 gives $Q_1 = 0$ directly. These are the only degree-dependent branches of the construction. On the invariant ring $K_0[u, v, W]$ the residual derivation is

$$(5.21) \quad D_T = (T\xi) \cdot \partial_\xi + S_T(u, v)\partial_W, \quad S_T(u, v) = \det(\xi, C_T(\xi, \xi)).$$

The rational functions in (5.16)–(5.18) select a K_0 -linear combination of the honest generators constructed above. They are not claimed to be coefficients of one rational vector field regular on the whole parameter space. This distinction lets us use the generic residual derivations without concealing any pole.

The output of Phase III is therefore algebraic rather than yet global: a finite collection of honest affine polynomial fields, together with the fact that its generic K_0 -span contains D_T for every $T \in \mathfrak{sl}_2(K_0)$. Phase IV globalizes the honest generators; it does not attempt to globalize the rational coefficients selecting D_T .

5.4. Phase IV: common twist and global extension. The preceding affine fields are not yet global on the projectivized parameter space or the Semple compactification. The product gauge first supplies the common parameter twist $\mathcal{O}(1) \boxtimes \mathcal{O}(1)$; the projective overlap and boundary calculations then supply the base twist $\mathcal{O}_{\mathbb{P}^2}(3)$. Together they give the honest global package used in residual rigidity.

Parameter and root gauges.

Lemma 5.2 (Product coefficient–root gauge). *Let a cone field have parameter multidegree $e = (e_1, e_2)$. Thus, under independent coefficient scalings $a_j \mapsto \lambda_j a_j$, its nonparameter coefficients scale by $\lambda_1^{e_1} \lambda_2^{e_2}$, whereas its coefficient vector q_i in the i -th block has one additional factor λ_i . Write its root component as $b_i w_i \partial_{w_i}$. On coefficient charts $\ell_i(a_i) = 1$, put*

$$(5.22) \quad c_i = \ell_i(q_i), \quad \tilde{q}_i = q_i - c_i a_i, \quad \tilde{b}_i = b_i - \frac{c_i}{d_i}.$$

If only the i -th coefficient chart changes, with transition ratio s_i and induced coordinate map Φ_i , then

$$(5.23) \quad \tilde{q}'_j = s_i^{e_i}(\Phi_i)_* \tilde{q}_j, \quad j = 1, 2,$$

$$(5.24) \quad \tilde{b}'_j = s_i^{e_i} \tilde{b}_j, \quad j \neq i,$$

$$(5.25) \quad \tilde{b}'_i = s_i^{e_i} \left(\tilde{b}_i + \frac{1}{d_i} \tilde{q}_i(\log s_i) \right).$$

Consequently the normalized representatives glue with parameter twist $\mathcal{O}_{\mathbb{P}(V_{d_1})}(e_1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(e_2)$.

Proof. The radial field R_i in (5.2) has coefficient component $d_i a_i$ and root component $w_i \partial_{w_i}$. Subtracting $(c_i/d_i)R_i$ therefore gives exactly (5.22); its coefficient component is tangent to $\ell_i(a_i) = 1$. On an overlap, the normalized coefficient representative is multiplied by $s_i^{e_i}$. The root changes by $w'_i = s_i^{1/d_i} w_i$, so its logarithmic derivative acquires $d_i^{-1} \tilde{q}_i(\log s_i)$. This gives (5.23)–(5.25). Performing the same calculation in the two blocks independently proves the product-twist statement. $\square$

When $d_i = 1$, the factors $1/d_i$ in the normalization and overlap formulas are simply one. Thus the triviality of the deck factor μ_1 removes a covering symmetry but does not remove the coefficient–root gauge.

The common parameter twist. When $d_1, d_2 \geq 2$, the reduced mixed family parametrized by (T, C, N_1, N_2) has parameter multidegree $(0, 0)$; no generator from the separate line-compatible package occurs. We multiply all these fields by $\ell_1 \ell_2$.

When $d_1 = 1$, Lemma 5.1 supplies the separate line-compatible package. Its T -, line-normal, and second-normal fields have multidegree $(0, 0)$, whereas its three kernel-acceleration fields are linear in the coefficients of the line and have multidegree $(1, 0)$. We multiply the former by $\ell_1 \ell_2$ and the latter by ℓ_2 . An i -th third-difference field has multidegree -1 in the i -th block and degree zero in the other block, so multiplication by $\ell_i^2 \ell_j$ ($j \neq i$) puts every summand in the common twist

$$(5.26) \quad \mathcal{O}_{\mathbb{P}(V_{d_1})}(1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(1).$$

At the generic incidence point the values of the ℓ_i are units. The Taylor-span identity (5.10) therefore selects, in each block of degree at least three, a linear combination of the twisted third differences whose cone coefficient is the negative of Q_i . A quadratic block contributes nothing because its reduced Q_i is zero, and at the residual combination the linear block contributes nothing by Lemma 5.1. Lemma 5.2 shows at the same time that the associated scalars c_i , the projective coefficient components, and the root-gauge corrections cancel. Thus the subtraction producing D_T is a linear combination of honest common-twist fields, rather than a rational field whose coefficients are asserted to be globally regular.

The base-chart pole bound. We next check the base twist. On a projective overlap let $\hat{z} = \phi(z)$ and let the i -th root frame change by $\hat{w}_i = g_i(z)w_i$; put $h_i = \log g_i$. The ordinary and root jets transform as

$$(5.27) \quad \hat{\xi} = D\phi \xi, \quad \hat{\eta} = D\phi \eta + D^2\phi(\xi, \xi),$$

$$(5.28) \quad \hat{\chi}_i = \chi_i + dh_i(\xi), \quad \hat{\psi}_i = \psi_i + dh_i(\eta) + d^2 h_i(\xi, \xi).$$

Varying these identities at a fixed base point and using (5.11), with $A = C(\xi, \xi)$, gives

$$(5.29) \quad \delta \hat{\xi} = D\phi(T\xi),$$

$$(5.30) \quad \delta \hat{\eta} = D\phi(T\eta + A) + 2D^2\phi(\xi, T\xi),$$

$$(5.31) \quad \delta \hat{\chi}_i = N_i(\xi) + dh_i(T\xi),$$

$$(5.32) \quad \delta \hat{\psi}_i = N_i(\eta) + dh_i(T\eta + A) + 2d^2 h_i(\xi, T\xi).$$

To read the denominators explicitly, take

$$s = x^{-1}, \quad r = y/x,$$

and write $(\alpha, \beta) = (s', r')$ and $(B_1, B_2) = (s'', r'')$. Differentiation of $x = s^{-1}$ and $y = rs^{-1}$ gives

$$(5.33) \quad \xi_x = -\frac{\alpha}{s^2}, \quad \xi_y = \frac{\beta}{s} - \frac{r\alpha}{s^2},$$

$$(5.34) \quad \eta_x = -\frac{B_1}{s^2} + \frac{2\alpha^2}{s^3}, \quad \eta_y = \frac{B_2}{s} - \frac{rB_1}{s^2} - \frac{2\alpha\beta}{s^2} + \frac{2r\alpha^2}{s^3}.$$

Thus a first jet has pole order at most two and a second jet has pole order at most three. For the acceleration part write $\widehat{A}_x = s^4 A_x$ and $\widehat{A}_y = s^4 A_y$; these are polynomial in the new first jets. Formula (5.30) gives the exact identities

$$(5.35) \quad \delta(s'') = -\frac{\widehat{A}_x}{s^2}, \quad \delta(r'') = \frac{\widehat{A}_y - r\widehat{A}_x}{s^3}.$$

For the standard root transition $h_i = -\log x = \log s$, the additional normal term is

$$(5.36) \quad dh_i(A) = -sA_x = -\frac{\widehat{A}_x}{s^3}.$$

For the matrix part, equations (5.33)–(5.34) give $N_i(\eta) = O(s^{-3})$. The entries of $D\phi$ have orders at least one in s , those of $D^2\phi$ have orders at least two, while $dh_i = O(s)$ and $d^2h_i = O(s^2)$. Hence

$$D\phi(T\eta), \quad D^2\phi(\xi, T\xi), \quad dh_i(T\eta), \quad d^2h_i(\xi, T\xi) = O(s^{-2}),$$

and the only possible third-order matrix/root term is $N_i(\eta)$. Equations (5.29)–(5.32) therefore prove, term by term, that the complete mixed field has base pole order at most three. Since every reduced Q_i has coefficient degree at most three, all components homogenize with $\mathcal{O}_{\mathbb{P}^2}(3)$. In the linear block the coefficients in (5.20) have z -degree at most two; the parameter-linear kernel fields have no additional z -dependence, and their second-block reductions again have degree at most three. Thus the degree-one generators introduce no further base twist.

The logarithmic corner. It remains to extend across logarithmic corners and the Semple fiber boundary. In a regular logarithmic frame write

$$\zeta = (x, y), \quad \theta = (X, Y), \quad \delta\zeta = T\zeta, \quad \delta\theta = T\theta + A(\zeta, \zeta),$$

where A has regular coefficients. This form is stable under a change of logarithmic frame: the second jet changes by a term linear in θ plus a term quadratic in ζ .

At a point of $C_i \cap C_j$, put

$$P_{ij}(z) = \begin{pmatrix} df_i|_z \\ df_j|_z \end{pmatrix}.$$

Transversality makes P_{ij} invertible in a neighborhood. The first and second root-jet equations solve there as

$$(5.37) \quad \xi = -P_{ij}^{-1} \begin{pmatrix} d_i w_i^{d_i} \chi_i \\ d_j w_j^{d_j} \chi_j \end{pmatrix},$$

$$(5.38) \quad \eta = -P_{ij}^{-1} \begin{pmatrix} d_i w_i^{d_i} \psi_i + d_i^2 w_i^{d_i} \chi_i^2 + \text{Hess}(f_i)(\xi, \xi) \\ d_j w_j^{d_j} \psi_j + d_j^2 w_j^{d_j} \chi_j^2 + \text{Hess}(f_j)(\xi, \xi) \end{pmatrix}.$$

Both expressions are holomorphic. Substitution in the mixed field shows that the induced T and A in the logarithmic frame remain regular at the corner.

The Semple boundary. Finally put

$$W = xY - yX, \quad p = y/x, \quad q = W/x^3, \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

A direct differentiation gives

$$(5.39) \quad \delta p = c + (d - a)p - bp^2,$$

$$(5.40) \quad \delta q = A_2(1, p) - pA_1(1, p) + (\text{tr } T - 3(a + bp))q.$$

In the reciprocal second-stage chart $r_2 = q^{-1}$ one has

$$(5.41) \quad \delta r_2 = -(A_2(1, p) - pA_1(1, p))r_2^2 - (\text{tr } T - 3(a + bp))r_2.$$

These expressions are polynomial. Interchanging x and y gives the other two charts, so the four standard charts cover the entire second Semple fiber boundary. Constant deck transformations

add constants to $\log w_i$ and therefore fix χ_i, ψ_i ; the root components are proportional to w_i , so both root boundary components are preserved.

Twist and role ledger. The four types of fields have different logical roles.

Reduced mixed field.: Before homogenization it has base pole at most 3 and preserves both tangency systems while changing the jet variables. Its parameter multidegree is $(0, 0)$, except for the line-compatible kernel-acceleration fields, which have multidegree $(1, 0)$.

Third difference in block i .: It has coefficient degree -1 in that block and base pole at most 3; it cancels an element of $\mathfrak{m}_z^3$. This field is absent when $d_i \leq 2$.

Homogenized field.: It has common parameter multidegree $(1, 1)$ and base pole at most 3; this is the honest global field on the universal two-component Semple tower.

Generic combination D_T .: Its coefficients lie in K_0 . It is used only to detect a generic normal direction and prove rigidity, and is not asserted to be global.

Thus every geometric differentiation is performed by an honest global twisted field. The rational functions in K_0 are used only to describe the span of their values at the generic point.

Proposition 5.3 (The mixed $\mathcal{O}(3)$ package). *Assume $d_1 \geq 1$ and $d_2 \geq 2$. On a nonempty Zariski-open subset of the universal family of ordered transverse pairs of degrees (d_1, d_2) , the finite constructions above define honest global logarithmic slanted vector fields on the relative second Semple tower with common twist*

$$(5.42) \quad \mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_1})}(1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(1).$$

Their span over the generic incidence field contains the three residual derivations D_T in (5.21), for $T \in \mathfrak{sl}_2$.

Proof. The ideal identities (5.13)–(5.15) prove tangency on every affine coefficient chart. Lemma 5.2 and (5.26) prove simultaneous descent through the two projective coefficient spaces and the two root gauges. Equations (5.27)–(5.36) give base pole order three, while (5.37)–(5.41) prove regularity at double logarithmic corners and on all Semple fiber charts. Indeed, the corner formulas express the ordinary jets polynomially in the two independent logarithmic root jets, and every root component has the boundary-preserving form $w_i \tilde{b}_i \partial_{w_i}$. The polynomial Semple formulas, together with those obtained by interchanging the two first-jet coordinates, cover the four standard charts.

The third-difference coefficients are monomials z^β with $|\beta| \leq 3$. Hence these fields, as well as the reduced mixed fields, have base pole order at most three and acquire no twist beyond $\mathcal{O}_{\mathbb{P}^2}(3)$. When both degrees are at least two, there are only finitely many constant data: three matrix directions T , six components of the symmetric map C , and four normal-coordinate directions in N_1, N_2 . When $d_1 = 1$, Lemma 5.1 replaces these by ten line-compatible mixed generators. In either case there are only finitely many third differences, none in a block of degree at most two. Their global sections therefore span a finite-dimensional complex vector space $\mathcal{V}$ with the common twist (5.42). The blockwise Taylor cancellation, performed after all summands have this common twist, says that the value of a suitable K_0 -linear combination of elements of $\mathcal{V}$ is D_T . It does not assert that the generic rational coefficients themselves define global vector fields. This is exactly the generic-span statement required later. $\square$

The finite-dimensional space $\mathcal{V}$ consists of genuine global twisted fields. At the generic incidence point its span contains D_H, D_E , and D_F . Consequently, if every element of $\mathcal{V}$ were tangent to the chosen prime divisor, then these three residual derivations would preserve its generic principal ideal. Phase V proves that this is impossible.

5.5. Phase V: residual curvature and rigidity. We now adapt the residual-rigidity computation of [8]. The invariant ring $K_0[u, v, W]$ has the same formal shape, but the W -components of the derivations are different: they contain the simultaneous solution of the two Hessian equations (5.17). The commutator calculation below verifies that this coupled correction still supplies the normal ∂_W direction needed for rigidity.

Choose coordinates over K_0 in which $g_1 = du$ and $g_2 = dv$, and write

$$K_1 = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad K_2 = \begin{pmatrix} d & e \\ e & f \end{pmatrix}.$$

For the standard basis

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

substitution in (5.18) gives

$$(5.43) \quad \begin{aligned} [D_H, D_E] - D_{[E, H]} &= -2v^2\Lambda(u, v)\partial_W, \\ [D_H, D_F] - D_{[F, H]} &= -2u^2\Lambda(u, v)\partial_W, \\ [D_E, D_F] - D_{[F, E]} &= -2uv\Lambda(u, v)\partial_W, \end{aligned}$$

where

$$\Lambda(u, v) = (a + e)u + (b + f)v.$$

This linear form is generically nonzero. When $d_1, d_2 \geq 2$, for example, the two local equations at $z = 0$

$$f_1 = 1 + z_1 + z_1^2, \quad f_2 = 1 + z_2 + z_2^2$$

give $K_1 = \text{diag}(1, 0)$, $K_2 = \text{diag}(0, 1)$, and $\Lambda = u + v$. When $d_1 = 1$ and $d_2 \geq 2$, use instead

$$f_1 = 1 + z_1, \quad f_2 = 1 + z_2 + z_2^2.$$

Then $K_1 = \text{diag}(-1, 0)$, $K_2 = \text{diag}(0, 1)$, and $\Lambda = -u + v \neq 0$. To formulate the generic conclusion without suppressing denominators, let Δ be a common denominator for the entries of G^{-1}, K_1, K_2 in the coordinate ring of $\mathcal{B}^\circ$. Then

$$\Delta\Lambda = \Lambda_1 u + \Lambda_2 v, \quad \Lambda_1, \Lambda_2 \in \mathcal{O}(\mathcal{B}^\circ).$$

The preceding specializations show that the pair (Λ_1, Λ_2) is not identically zero. Because $\mathcal{B}^\circ$ is irreducible, the locus

$$(5.44) \quad \mathcal{B}_\Lambda^\circ = \{\Delta\Lambda_1 \neq 0\} \cup \{\Delta\Lambda_2 \neq 0\}$$

is a nonempty Zariski-open subset, and $\Lambda \neq 0$ in $K_0[u, v]$. After clearing the joint variables z , the assertion that both numerator coefficients vanish identically is a closed condition on the two parameters. Since it is not the whole parameter space, its complement is a nonempty parameter open. Intersecting with the dense smooth transverse locus gives the open subset of $\mathcal{S}_{d_1, d_2}^{\text{snc}}$ used below.

Proposition 5.4 (Residual rigidity). *Let L/K_0 be a characteristic-zero extension. There is no nonconstant weighted-homogeneous polynomial $P \in L[u, v, W]$ of positive weight whose principal ideal is preserved by D_H, D_E, D_F .*

Proof. The ideal (P) is then preserved by commutators and linear combinations. Equation (5.43) supplies a nonzero derivation $S(u, v)\partial_W$ preserving (P) . If $\deg_W P > 0$, the equality $S\partial_W P = AP$ is impossible by comparison of the W -degrees unless $A = 0$; since the polynomial ring is a domain, $\partial_W P = 0$. The same conclusion is immediate when $\deg_W P = 0$. Hence $P \in L[u, v]$.

The W -parts of the three D_T now disappear, so the full binary $\mathfrak{sl}_2$ -action preserves (P) . In a fixed ordinary degree the quotients are scalars. The two nilpotent operators $v\partial_u$ and $u\partial_v$ can have only the scalar zero, whence $vP_u = uP_v = 0$. Therefore P is constant, contradicting its positive weight. $\square$

Consequently the generic prime divisor cannot be tangent to every honest field in the mixed package: otherwise its principal ideal would be preserved by all three residual directions, contrary to Proposition 5.4. Phase VI records the natural action of these invariant fields and restricts the resulting global derivative to each parameter fibre.

5.6. Phase VI: the natural global action and the second equation. The mixed package supplies invariant twisted vector fields on the universal vertical two-jet space. A relative invariant jet differential is represented there by an equivariant polynomial in the jet variables. Consequently an honest field acts on it directly as a derivation; the result is again a global equivariant invariant jet polynomial, with the twist of the field added to the original twist. This is Siu's standard mechanism for producing the second jet differential. The remaining logic is

$$\begin{aligned} \text{generic span of honest fields} &\implies \text{rigidity forbids common tangency,} \\ &\implies \mathcal{L}_V f \notin (f) \implies \omega_{2,a}. \end{aligned}$$

The first implication uses the generic prime equation. The last is an immediate consequence of the following natural-action lemma, rather than a cohomological lift from the divisor.

Lemma 5.5 (Natural action on equivariant jet polynomials). *Let $\mathcal{J} \rightarrow B$ be a universal vertical two-jet space, let $\mathcal{L}$ and $\mathcal{M}$ be the homogeneous line bundles determined by the base and parameter gradings, and let f be an invariant equivariant jet-polynomial section of $\mathcal{L}$. If*

$$\mathcal{V} \in H^0(\mathcal{J}, T_{\mathcal{J}} \otimes \mathcal{M})$$

is an invariant slanted field whose homogeneous local expressions descend on all projective and Semple overlaps, then

$$\mathcal{L}_V f := \mathcal{V}(f) \in H^0(\mathcal{J}, \mathcal{L} \otimes \mathcal{M})$$

is a global invariant equivariant jet-polynomial section. After restriction to a parameter fibre, the parameter factor of $\mathcal{M}$ is a constant one-dimensional space, while its base twist remains unchanged.

Proof. Work first on the product of the affine coefficient cones and the affine jet cones. There f is an equivariant polynomial function and $\mathcal{V}$ is a polynomial derivation, so $\mathcal{V}(f)$ is defined without choosing a connection. The coefficient-root gauge identities, the projective overlap formulas, and the Semple-chart formulas proved in Phase IV show that this derivative has the product homogeneity $\mathcal{L} \otimes \mathcal{M}$ on every overlap. Hence it descends globally. Since both f and $\mathcal{V}$ are invariant under source reparametrization, so is their Lie derivative. Restricting this already global relative section to a fibre gives the last assertion. $\square$

Lemma 5.6 (Generic equation and preservation of the prime ideal). *Let $\mathcal{Z} \subset \mathcal{X}_2$ be the relative integral horizontal divisor supplied by Proposition 4.3, and assume $b_2 > 0$. On the dense incidence chart used in (5.6), after a characteristic-zero algebraic field extension L/K_0 , the affine cone over the generic fiber of $\mathcal{Z}$ is defined by an irreducible weighted-homogeneous polynomial*

$$P \in L[u, v, W]$$

of positive weight. Let $\mathcal{V}$ be an honest twisted slanted field from Proposition 5.3. After rationally trivializing its common twist, let $\tilde{V}$ be the induced derivation. Here $\mathcal{L}_V f$ denotes the natural global action of Lemma 5.5. Then

$$(5.45) \quad (\mathcal{L}_V f)|_{\mathcal{Z}} = 0 \text{ at the generic point of } \mathcal{Z} \iff \tilde{V}(P) \in (P).$$

Consequently, if every global derivative vanishes generically along $\mathcal{Z}$, then each of D_H, D_E, D_F preserves the principal prime ideal (P) .

Proof. The divisor $\mathcal{Z}$ is integral. If it were contained in the complement of the chart $f_1 f_2 \neq 0$ or in the Semple boundary, it would equal one of those irreducible boundary divisors. This is impossible by classes: a base boundary has positive class $d_i h$, the second Semple boundary has class $-u_1 + u_2$, whereas $[\mathcal{Z}] = b_1 u_1 + b_2 u_2 - t h$ with $t > 0$ and $b_1 \geq 2b_2 > 0$. Thus the displayed chart meets the generic point of $\mathcal{Z}$. More generally, $b_2 > 0$ implies that $\mathcal{Z}$ dominates X_1 and hence $\mathbb{P}^2$; it therefore cannot be contained in the pullback of the determinant divisor $g_1 \wedge g_2 = 0$, an affine transition divisor, or a chosen Semple-chart boundary. The coefficient-chart linear forms are units at the generic parameter after the corresponding affine parameter chart is

chosen. This accounts for every denominator listed in (5.8). The invariant affine two-jet cone over the generic incidence point has coordinate ring $L[u, v, W]$ by (5.6). The pullback of the generic point of $\mathcal{Z}$ is a height-one prime in this unique factorization domain. It is therefore generated by an irreducible polynomial P . The prime is stable under the weighted $\mathbb{G}_m$ action. Since the units of $L[u, v, W]$ are the elements of L^* , its generator is a semi-invariant for this action and may be chosen weighted-homogeneous. Its weight is positive because all three variables have positive weight, while a nonzero weight-zero element of $L[u, v, W]$ is a unit and cannot generate a height-one prime. Every derivation of K_0 extends uniquely through the algebraic extension L/K_0 in characteristic zero, so all the residual fields act on this same ring.

At the generic point, choose rational frames of $\mathcal{L}$ and of the common twisting line. Up to a unit e , the relative section then has the form $f = eP$. Hence

$$\tilde{V}(eP) = e\tilde{V}(P) + \tilde{V}(e)P.$$

Reduction modulo (P) proves (5.45); it also shows that the assertion is independent of both rational frames. The set of derivations preserving a fixed ideal is a vector space over the generic incidence field. Proposition 5.3 says that the generic span of the honest fields contains D_H, D_E, D_F , which proves the last assertion. $\square$

Proposition 5.7 ($\mathcal{O}(3)$ slanted derivative). *After shrinking the parameter open in Proposition 4.3, write the irreducible reduced horizontal divisor of $\omega_{1,a}$ as*

$$Z_a \sim b_1 u_1 + b_2 u_2 - th, \quad b_1 \geq 2b_2 > 0, \quad m = b_1 + b_2.$$

After a further nonempty Zariski-open shrinking, for every remaining a there is

$$\omega_{2,a} \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t+3))$$

such that the common zero set of $\omega_{1,a}$ and $\omega_{2,a}$ has codimension at least two away from $\Gamma_{2,a}$.

Proof. The open-set bookkeeping has three steps: the factor family and the global field package are defined on the first open; the curvature form Λ is nonzero on the second; and the last shrinking below makes one global derivative cut the residual divisor properly on every remaining fibre. Each condition defines a nonempty Zariski-open subset, so their finite intersection is nonempty.

Use the relative notation of Proposition 4.3. On the parameter open under consideration set

$$\mathcal{A} = \mathcal{O}_{\mathcal{S}}(1, 1)|_{\mathcal{U}}, \quad \mathcal{M} = \Pi^* \text{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(3) \otimes p_{\mathcal{U}}^* \mathcal{A}.$$

Every honest field $\mathcal{V}$ in Proposition 5.3 is a section of $T_{\mathcal{X}_2} \otimes \mathcal{M}$; in particular, it is allowed to have a component in the parameter directions. By Lemma 5.5, it acts on the equivariant relative equation and defines the global relative section

$$\mathcal{L}_{\mathcal{V}} f \in H^0(\mathcal{X}_2, \mathcal{L} \otimes \mathcal{M}).$$

If all their restrictions vanished at the generic point of $\mathcal{Z}$, then Lemma 5.6 would give a positive-weight irreducible $P \in L[u, v, W]$ whose ideal is preserved by D_H, D_E, D_F . This contradicts Proposition 5.4. Hence some honest field $\mathcal{V}$ satisfies

$$(5.46) \quad (\mathcal{L}_{\mathcal{V}} f)|_{\mathcal{Z}} \neq 0 \quad \text{at the generic point of } \mathcal{Z}.$$

The section in (5.46) is already defined on the entire relative tower. Its restriction to the generic fibre cuts the residual divisor properly. Properness of $\mathcal{Z} \rightarrow \mathcal{U}$ and cohomology and base change show that the locus where the fibrewise restriction vanishes identically along Z_a is closed. After one final shrinking it is therefore nonzero on Z_a for every remaining parameter. Trivializing the one-dimensional factor $\mathcal{A}|_a$ by a nonzero scalar gives directly

$$s_{2,a} := (\mathcal{L}_{\mathcal{V}} f)|_a \in H^0(X_{2,a}, L_a \otimes M_a), \quad \begin{aligned} L_a &= \mathcal{O}_{X_{2,a}}(b_1, b_2) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-t), \\ M_a &= \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(3), \end{aligned}$$

with $s_{2,a}|_{Z_a} \neq 0$. No extension from Z_a and no auxiliary cohomological lifting are involved.

Let γ_a be the canonical section of $\mathcal{O}_{X_{2,a}}(\Gamma_{2,a}) = \mathcal{O}_{X_{2,a}}(-1, 1)$. Since

$$L_a \otimes \mathcal{O}_{X_{2,a}}(b_1 \Gamma_{2,a}) = \mathcal{O}_{X_{2,a}}(m) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

the sections

$$\omega_{1,a} = \gamma_a^{b_1} s_a, \quad \omega_{2,a} = \gamma_a^{b_1} s_{2,a}$$

have the same weighted degree m and twists $-t$ and $-t+3$. The direct-image formula identifies the latter with the asserted invariant jet differential. Finally, $s_{2,a}|_{Z_a} \neq 0$, so the two sections have no common divisorial component outside $\Gamma_{2,a}$. $\square$

Corollary 5.8 ($\mathcal{O}(3)$ slanted-field alternative). *Let $f : \mathbb{C} \rightarrow \mathbb{P}^2$ be algebraically nondegenerate and suppose $t > 3$ and $f^* \omega_{1,a} \equiv 0$. Then either $f^* \omega_{2,a} \neq 0$ for the differential in Proposition 5.7, or the first logarithmic lift $f_{[1]}$ is algebraically degenerate.*

Proof. Assume that both pullbacks vanish, and let $Y = \overline{f_{[2]}(\mathbb{C})}^{\text{Zar}} \subset X_2$. By Lemma 2.4, Y is not contained in Γ_2 . Hence $Y \setminus \Gamma_2$ is dense in Y and lies in the codimension-two common zero set of $\omega_{1,a}$ and $\omega_{2,a}$. Since $\dim X_2 = 4$, it follows that $\dim Y \leq 2$. Therefore $\pi_{2,1}(Y)$ is a proper algebraic subset of the threefold X_1 and contains the image of $f_{[1]}$. $\square$

6. THE TWO COMPLEMENTARY MECHANISMS

The argument now follows a fixed decision tree. If the chosen factor does not vanish on the lifted curve, its original twist gives the estimate. If it vanishes, the case $b_2 = 0$ is handled on X_1 ; for $b_2 > 0$, a pole-three derivative treats $t > 3$, while the key vanishing lemma and the zero-locus theorem treat $t \leq 3$.

For one of the six boundary pairs in Lemma 1.2, let $\mathcal{U}_{\text{key}} = \mathcal{U}_{\text{van}}(d_1, d_2)$; for every other degree pair in our range, put $\mathcal{U}_{\text{key}} = \mathcal{S}_{d_1, d_2}^{\text{unc}}$. Let $\mathcal{U}_0$ be the intersection of $\mathcal{U}_{\text{key}}$ with the parameter open furnished by Proposition 4.3. It is nonempty because the parameter space is irreducible. Work first over $a \in \mathcal{U}_0$, and put $D = D_a$ and $\omega_1 = \omega_{1,a}$. Let $f : \mathbb{C} \rightarrow \mathbb{P}^2$ be algebraically nondegenerate and suppose $f(\mathbb{C}) \not\subset D$. If $f^* \omega_1 \neq 0$, Theorem 2.5 and $m/t \leq \rho_0^{-1}$ give the required estimate directly. It remains to treat the vanishing case

$$(6.1) \quad f^* \omega_1 \equiv 0.$$

Write the selected horizontal component as

$$Z \sim b_1 u_1 + b_2 u_2 - th.$$

If $b_2 = 0$, then $Z = \pi_{2,1}^{-1}(Z_1)$ for a proper divisor $Z_1 \subset X_1$. By Lemma 2.4, the regular lift is not contained in Γ_2 . Hence (6.1) implies

$$f_{[1]}(\mathbb{C}) \subset Z_1,$$

so $f_{[1]}$ is algebraically degenerate and Theorem 2.6 applies. This branch uses no slanted field. Hence in the two cases below we may assume $b_2 > 0$. Proposition 5.7, applied after restricting the factor family to $\mathcal{U}_0$, supplies a nonempty open subset $\mathcal{U}_1 \subset \mathcal{U}_0$ on which the chosen differential and its slanted derivative are defined simultaneously. In the remainder of this section we work over $a \in \mathcal{U}_1$.

6.1. Case 1: $t > 3$. Corollary 5.8 supplies an independent

$$\omega_{2,a} \in H^0(\mathbb{P}^2, E_{2,m} \bar{T}^* \otimes \mathcal{O}(-t+3)).$$

If $f^* \omega_{2,a} \neq 0$, Theorem 2.5 gives

$$T_f(r) \leq \frac{m}{t-3} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

Since $m/t \leq \rho_0^{-1}$ and $t \geq 4$,

$$(6.2) \quad \frac{m}{t-3} \leq \frac{1}{\rho_0} \frac{t}{t-3} \leq \frac{4}{\rho_0}.$$

If $f^* \omega_{2,a} \equiv 0$, Corollary 5.8 makes $f_{[1]}$ algebraically degenerate, and Theorem 2.6 applies.

6.2. Case 2: $t \leq 3$. Since $b_2 > 0$ and $b_1 \geq 2b_2$, one has $m \geq 3$. If $\rho_{\text{DEG}} > 1$, then automatically $t/m \leq 1 < \rho_{\text{DEG}}$. Otherwise the degree pair is one of the six boundary pairs, and Lemma 1.2 implies that the existing nonzero ω_1 cannot have $t/m \geq \rho_{\text{DEG}}$. Thus in every case

$$(6.3) \quad \rho = \frac{t}{m} < \rho_{\text{DEG}}(d_1, d_2).$$

Theorem 2.11 applies. For every rational $0 < \tau < \min\{3, \tau_1(t/m)\}$ it gives either

$$T_f(r) \leq \frac{3}{\tau} N_f^{[1]}(r, D) + o(T_f(r)) \parallel$$

or the algebraic degeneracy of $f_{[1]}$, followed by (2.9).

6.3. The effective constant. The discrete gap below the zero-locus threshold is explicit. Set

$$(6.4) \quad \rho_* = \max_{1 \leq q \leq 3} \frac{q}{\max\{3, \lfloor q/\rho_{\text{DEG}} \rfloor + 1\}}, \quad \tau_* = \min\{3, \tau_1(\rho_*)\}.$$

For integers $1 \leq t \leq 3$, $m \geq 3$ satisfying $t/m < \rho_{\text{DEG}}$, one has $t/m \leq \rho_*$. The function $3/\tau_1(\rho)$ is increasing on the relevant interval. Consequently any integer strictly larger than

$$(6.5) \quad \max \left\{ \rho_0^{-1}, \frac{4}{\rho_0}, \frac{3}{\tau_*}, \frac{1}{\kappa} \right\}$$

is an admissible $\mathcal{A}_{d_1, d_2}$ in Theorem 1.3. In the order displayed, the four entries account for the original section, its pole-three derivative, the zero-locus section, and McQuillan's branch.

6.4. The sharper conic–quintic dichotomy. For $(d_1, d_2) = (2, 5)$ one has

$$\kappa = 4, \quad \bar{c}_1^2 = 16, \quad \bar{c}_2 = 21, \quad \rho_{\text{DEG}} = \frac{19}{48}.$$

The extra case $(m, t) = (8, 3)$ in $\mathfrak{F}_{45}(2, 5)$ changes the useful split from $3/8$ to $1/3$. Put $\rho = t/m$ and assume $b_2 > 0$.

If $\rho \leq 1/3$, then $\rho < 19/48$, so Theorem 2.11 applies. Since $3/\tau_1(\rho)$ is increasing on this interval, it is enough to evaluate the endpoint. Formula (2.13) gives

$$(6.6) \quad \tau_1(1/3) = \frac{45 - \sqrt{1989}}{6}, \quad \frac{3}{\tau_1(1/3)} = \frac{45 + \sqrt{1989}}{2} < 45.$$

The last inequality follows from $1989 < 45^2$. For the actual ratio $\rho \leq 1/3$, one may therefore choose a rational number $1/15 < \tau < \tau_1(\rho)$; the zero-locus coefficient $3/\tau$ is then strictly smaller than 45.

Suppose instead that $\rho > 1/3$. If $t \leq 3$, then $m < 3t$ and hence $(m, t) \in \mathfrak{F}_{45}(2, 5)$, contradicting the nonzero choice of ω_1 and Lemma 1.2. Thus $t \geq 4$, and the pole-three derivative is available. Integrality gives $m \leq 3t - 1$, whence

$$(6.7) \quad \frac{m}{t-3} \leq \frac{3t-1}{t-3} = 3 + \frac{8}{t-3} \leq 11.$$

If the differentiated equation vanishes as well, the first lift is algebraically degenerate and McQuillan's estimate applies.

Finally,

$$(6.8) \quad \rho_{\text{RR}}(2, 5) = \frac{2}{9} \left(8 - \sqrt{\frac{455}{8}} \right) > \frac{1}{10};$$

indeed, after squaring, the last inequality is $22750 < 22801$. Choose a rational number

$$\frac{1}{10} < \rho_0 < \rho_{\text{RR}}(2, 5).$$

The direct branch then has coefficient $\rho_0^{-1} < 10$, (6.6) bounds the zero-locus branch by a number strictly smaller than 45, and (6.7) bounds the differentiated branch by 11. McQuillan's coefficient is $1/4$. Therefore the integer 45 covers every branch for a conic and a quintic.

6.5. The explicit line–octic constant. For $(d_1, d_2) = (1, 8)$ one has $\kappa = 6$ and $\rho_{\text{DEG}} = 3/8$. Formula (6.4) therefore gives $\rho_* = 1/3$. Substitution in (2.13) yields

$$(6.9) \quad \tau_1(1/3) = \frac{23 - 5\sqrt{21}}{2}, \quad \frac{3}{\tau_1(1/3)} = \frac{3}{2}(23 + 5\sqrt{21}) < 69.$$

Moreover,

$$(6.10) \quad \rho_{\text{RR}}(1, 8) = \frac{1}{3} \left(8 - \sqrt{\frac{119}{2}} \right) > \frac{2}{23};$$

indeed, after squaring, it reduces to $62951 < 63368$. Choose a rational number

$$\frac{2}{23} < \rho_0 < \rho_{\text{RR}}(1, 8).$$

Then the direct branch has coefficient $\rho_0^{-1} < 23/2$, the differentiated branch has coefficient $4/\rho_0 < 46 < 69$, and (6.9) permits, for the actual ratio, a rational choice $3/69 < \tau < \tau_1(\rho)$; hence the zero-locus coefficient is strictly below 69. McQuillan's coefficient is $1/6$. Thus the integer 69 covers every branch for a line and an octic.

6.6. The sharper ratio dichotomy for two cubics. For $(d_1, d_2) = (3, 3)$, the stronger part of Lemma 1.2 improves the preceding general estimate. Put $\rho = t/m$ and retain the assumption $b_2 > 0$.

Suppose first that $\rho \leq 1/5$. Since $1/5 < \rho_{\text{DEG}}(3, 3) = 1/4$, Theorem 2.11 applies, irrespective of the absolute size of t . The function $3/\tau_1(\rho)$ is increasing on $[0, 1/4)$, and

$$(6.11) \quad \tau_1(1/5) = \frac{57 - \sqrt{3189}}{10}, \quad \frac{3}{\tau_1(1/5)} = \frac{57 + \sqrt{3189}}{2} \approx 56.73561582 < 57.$$

Thus one can choose a rational number $3/57 < \tau < \tau_1(\rho)$, and the zero-locus branch has coefficient strictly smaller than 57.

Suppose next that $\rho > 1/5$. If $t \leq 3$, integrality gives $m \leq 5t - 1$, contradicting the cubic finite vanishing in Lemma 1.2. Hence $t \geq 4$, so the mixed $\mathcal{O}(3)$ derivative is available. Since $m \leq 5t - 1$,

$$(6.12) \quad \frac{m}{t-3} \leq \frac{5t-1}{t-3} \leq 19.$$

If the differentiated equation also vanishes on the lifted curve, Corollary 5.8 makes the first lift algebraically degenerate, and Theorem 2.6 applies. Consequently the two branches under $f^*\omega_1 \equiv 0$ and $b_2 > 0$ are bounded by 57; the direct and $b_2 = 0$ branches are included in the final assembly.

7. PROOFS OF THE MAIN STATEMENTS

We now assemble the preceding alternatives. No new numerical or geometric input is needed at this stage.

Proof of Theorem 1.3. The fixed pair (M, T) of Proposition 4.1 works throughout $\mathcal{S}_{d_1, d_2}^{\text{nc}}$. Intersect the parameter open of Proposition 4.3 with $\mathcal{U}_{\text{key}}$ and call the resulting nonempty open set $\mathcal{U}_0$. The integers b_1, b_2, m, t are fixed on this family. If $b_2 = 0$, put $\mathcal{U}_{\text{SMT}}(d_1, d_2) = \mathcal{U}_0$. If $b_2 > 0$, apply Proposition 5.7 to the restricted family and let $\mathcal{U}_{\text{SMT}}(d_1, d_2) \subset \mathcal{U}_0$ be the nonempty open set it provides.

Fix $a \in \mathcal{U}_{\text{SMT}}(d_1, d_2)$ and choose $\omega_{1,a}$ as in Lemma 4.2 and Proposition 4.3. If $f^*\omega_{1,a} \not\equiv 0$, the jet Second Main Theorem gives the estimate with constant at most ρ_0^{-1} . Under (6.1), the branch $b_2 = 0$ makes $f_{[1]}$ algebraically degenerate and is handled directly by Theorem 2.6.

Suppose therefore that $b_2 > 0$. For a degree pair other than $(3, 3)$, Case 1 applies when $t > 3$ and gives (6.2) or McQuillan's stronger estimate. When $t \leq 3$, Case 2 and (6.4) give the zero-locus estimate or McQuillan's estimate. Hence any integer strictly larger than (6.5) is admissible.

For $(d_1, d_2) = (1, 8)$ choose ρ_0 as in (6.10); the preceding line–octic calculation shows that the integer 69 is admissible.

For $(d_1, d_2) = (2, 5)$ choose ρ_0 as in (6.8); use the sharper $1/3$ dichotomy of the preceding conic–quintic subsection in place of the coarser general split. The direct, zero-locus, differentiated, and algebraically degenerate branches are then bounded respectively by 10, 45, 11, and $1/4$, with strict inequality where needed. Hence the integer 45 is admissible.

For $(d_1, d_2) = (3, 3)$ choose $1/16 < \rho_0 < \rho_{\text{RR}}(3, 3)$, so the direct branch has coefficient less than 16. Under (6.1), use instead the sharper ratio dichotomy of the preceding subsection: equation (6.11) gives a coefficient less than 57 when $t/m \leq 1/5$, and (6.12) gives a coefficient at most 19 when $t/m > 1/5$. Thus the integer 57 works for two cubics. This proves (1.2). $\square$

Proof of Corollary 1.4. Fix

$$a \in \mathcal{U}_{\text{SMT}}(d_1, d_2) \cap \mathcal{S}_{\text{Chen}}(d_1, d_2)$$

and let

$$f : \mathbb{C} \longrightarrow \mathbb{P}^2 \setminus D_a.$$

If f is algebraically nondegenerate, then $N_f^{[1]}(r, D_a) = 0$, and Theorem 1.3 gives $T_f(r) \leq o(T_f(r))$, impossible for a nonconstant curve. If f is algebraically degenerate, Corollary 2.14 makes it constant. Hence the complement is Brody hyperbolic.

It remains to pass from Brody hyperbolicity to the asserted metric statement. The one-dimensional boundary strata are

$$C_1 \setminus C_2, \quad C_2 \setminus C_1.$$

Since the curves meet transversally in $d_1 d_2$ points, their logarithmic Euler characteristics satisfy

$$2g(C_i) - 2 + d_1 d_2 > 0 \quad (i = 1, 2)$$

throughout the stated degree range. Thus both punctured curves are Brody hyperbolic, and the zero-dimensional stratum $C_1 \cap C_2$ is finite. We use the following form of Green's criterion: if X is projective, $D = \bigcup_{i=1}^r D_i$ is a simple-normal-crossing divisor, and both $X \setminus D$ and every open stratum

$$\bigcap_{i \in I} D_i \setminus \bigcup_{j \notin I} D_j \quad (\emptyset \neq I \subset \{1, \dots, r\})$$

are Brody hyperbolic, then $X \setminus D$ is hyperbolically embedded in X ; see [9] and the precise restatement [10, Theorem 4.1]. The strata in the present case are exactly the complement, the two punctured curves above, and the finite set $C_1 \cap C_2$. The criterion therefore shows that the complement is hyperbolically embedded in $\mathbb{P}^2$; in particular it is Kobayashi hyperbolic. The set $\mathcal{U}_{\text{SMT}}(d_1, d_2) \cap \mathcal{S}_{\text{Chen}}(d_1, d_2)$ is very general by the discussion following Corollary 1.4. $\square$

8. PROOF OF THE FINITE KEY VANISHING LEMMA: EXACT FINITE COMPUTATION

We now prove Lemma 1.2. It is enough to establish the vanishing at one transverse pair, because upper semicontinuity will then give the required nonempty parameter open. After multiplication by a nonzero homogeneous polynomial reduces to the least twist for each m , the finite range is

$$H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}(-t)) = 0$$

for the pairs listed in the table following Lemma 1.2. The guiding principle is to generate candidate two-jet differentials on one large ordinary chart, test their logarithmic regularity on one mixed logarithmic chart, and impose the available symmetries before constructing any matrix. This chart-and-divisibility strategy starts from the one-component normal form in [6, Proposition 3.1 and Section 3]. The argument below is not a restatement of that calculation:

two defining polynomials produce two residue directions, their projective weights must be balanced when $d_1 \neq d_2$, and the overlap equations are coupled. For $(3, 3)$ we impose the $(\mu_3)^2$ action and coordinate exchange before constructing the matrices. The pairs $(2, e)$, $e = 5, 6$, have a different cyclic action with coordinate exchange. The line pairs $(1, e)$, $e = 8, 9$, retain the cyclic action but not the involution, while $(3, 4)$ is small enough to run as one unsplit block. The cubic organization is related to the symmetry method developed in [8]. That citation concerns the organizational idea of splitting a finite calculation by symmetry; the two-component normal form, divisibility criterion, matrices, and verification program used here are derived independently in this section. All operations are performed with integral polynomials. Every characteristic-zero conclusion is certified by full column rank at an explicitly recorded good prime; a second prime is used as an independent reproducibility check whenever practical.

The proof has four stages. We first choose explicit SNC test pairs. We then derive a finite ambient normal form and translate regularity on the mixed chart into divisibility equations. Next we decompose these equations by the available finite symmetries. Finally, exact modular row reduction certifies that every resulting block has zero kernel.

8.1. Degree-uniform test pairs. We use the following integral pairs; in each row, $A = 0$ and $B = 0$ are the two components of the test divisor:

$$(8.1) \quad \begin{array}{c|cc} (d_1, d_2) & A & B \\ \hline (3, 3) & X^3 + 2Y^3 + Z^3 & 2X^3 + Y^3 + Z^3 \\ (3, 4) & X^3 + Y^3 + Z^3 & X^4 + Y^4 + Z^4 \\ (2, e), e = 5, 6 & XY + Z^2 & X^e + Y^e + Z^e \\ (1, 8) & Y & X^8 + Y^8 + Z^8 + X^4Y^4 \\ (1, 9) & Y & X^9 + Y^9 + Z^9 + X^3Y^3Z^3. \end{array}$$

Each row gives an SNC divisor and remains SNC after adjoining the line $L = \{Z = 0\}$. For $(3, 4)$, proportional gradients at a common point would force all nonzero coordinates to be equal; neither two nor three equal nonzero coordinates can satisfy the cubic equation. On L , the two equations would give simultaneously $(X/Y)^3 = -1$ and $(X/Y)^4 = -1$, which is impossible; each curve meets L transversally. The assertions for $(3, 3)$ are proved below.

For the rows $(2, e)$, the conic $A = XY + Z^2$ and the Fermat curve $B_e = X^e + Y^e + Z^e$ are smooth. Suppose that they meet nontransversally. If $Z = 0$, then $A = 0$ forces $XY = 0$, which is incompatible with $B_e = 0$. Thus we may scale $Z = 1$, so $XY = -1$. Writing $\nabla B_e = \lambda \nabla A$, the third component gives $\lambda = e/2$, and the first two components give

$$2X^{e-1} = Y, \quad 2Y^{e-1} = X.$$

Multiplication by X and Y , respectively, yields $X^e = Y^e = -1/2$. Hence

$$(XY)^e = (-1)^e = X^e Y^e = \frac{1}{4},$$

a contradiction for $e = 5, 6$. On L , the conic meets the line at $[1 : 0 : 0]$ and $[0 : 1 : 0]$, neither of which lies on B_e ; the Fermat curve also meets L transversally. Thus $A + B_e + L$ is SNC.

For $(1, 8)$, let

$$B_8 = X^8 + Y^8 + Z^8 + X^4Y^4.$$

At a singular point of B_8 , the equation $(B_8)_Z = 8Z^7 = 0$ gives $Z = 0$. Neither X nor Y can vanish, while the other two partial derivatives give

$$2X^4 + Y^4 = 0, \quad X^4 + 2Y^4 = 0.$$

Their determinant is 3, a contradiction. Thus B_8 is smooth. On the line $A = \{Y = 0\}$ its intersection equation is $X^8 + Z^8 = 0$, so X and Z are nonzero and the intersection is transverse. Smoothness likewise implies transversality with $L = \{Z = 0\}$, because $(B_8)_Z$ vanishes there. Finally, $A \cap L = [1 : 0 : 0] \notin B_8$, and hence $A + B_8 + L$ is SNC.

For $(1, 9)$, put

$$B_9 = X^9 + Y^9 + Z^9 + X^3Y^3Z^3.$$

A singular point with a zero coordinate is impossible from the remaining partial derivatives. If all coordinates are nonzero, set $a = X^3$, $b = Y^3$, and $c = Z^3$. The three singularity equations become

$$3a^2 + bc = 3b^2 + ac = 3c^2 + ab = 0.$$

Multiplication gives $27 = -1$, again a contradiction. Hence B_9 is smooth. Its intersections with A and L are transverse by the same restriction argument, and $A \cap L = [1 : 0 : 0] \notin B_9$. Thus $A + B_9 + L$ is SNC.

8.2. A special symmetric pair of cubic curves. We work with the following special transverse pair:

$$(8.2) \quad A = X^3 + 2Y^3 + Z^3, \quad B = 2X^3 + Y^3 + Z^3.$$

Let

$$D = \{AB = 0\}.$$

Both cubics are smooth, and the two curves meet transversally. Indeed, if

$$A = B = 0,$$

then

$$X^3 = Y^3, \quad 3X^3 + Z^3 = 0.$$

Thus all intersection points have nonzero coordinates. The two gradients cannot be proportional: their Z -components would force the scalar to be 1, whereas the X -components would then require $3X^2 = 6X^2$. Moreover, $A = B = Z = 0$ has no projective solution, and each cubic meets $L = \{Z = 0\}$ transversally. Hence $A + B + L$ is SNC.

The pair has two useful symmetries. First, the involution

$$(8.3) \quad \sigma(X, Y, Z) = (Y, X, Z)$$

satisfies

$$\sigma^*A = B, \quad \sigma^*B = A.$$

Second, for $\alpha^3 = \beta^3 = 1$, the diagonal transformation

$$(8.4) \quad [X : Y : Z] \mapsto [\alpha X : \beta Y : Z]$$

fixes both A and B . Thus the finite-dimensional spaces appearing in the computation decompose into characters of $(\mu_3)^2$, and the involution σ exchanges the two character indices.

For the pairs $(2, e)$ in (8.1), the corresponding symmetry is

$$[X : Y : Z] \mapsto [\zeta X : \zeta^{-1}Y : Z], \quad \zeta^e = 1,$$

together with the involution $X \leftrightarrow Y$. The monomial

$$x^r y^s (x')^i (y')^j W_{xy}^k$$

has character $r - s + i - j \pmod{e}$, because W_{xy} has character zero. Coordinate exchange sends the character c to $-c$ and multiplies the k -th Wronskian layer by $(-1)^k$. Consequently the $e = 5$ candidate space splits into six blocks indexed by $\{0\}, \{1, 4\}, \{2, 3\}$ and the two involution signs; for $e = 6$ it splits into eight blocks indexed by $\{0\}, \{1, 5\}, \{2, 4\}, \{3\}$ and the two signs.

For either line pair $(1, e)$ the same cyclic transformation

$$[X : Y : Z] \mapsto [\zeta X : \zeta^{-1}Y : Z], \quad \zeta^e = 1,$$

preserves both components. There is no coordinate-exchange involution, because the distinguished component is the line $Y = 0$. Since $a = y$ has character -1 on the Z -chart and b is invariant, a source monomial in the k -th Wronskian layer has the shifted character

$$(8.5) \quad r - s + i - j + n_k \pmod{e}, \quad n_k = m - 2k.$$

Thus the complete candidate space splits into exactly e cyclic character blocks. The shift by n_k is essential: it is contributed by the factor a^{-n_k} in the normal form.

8.3. The ordinary and mixed logarithmic charts. The ordinary chart below supports the finite ansatz and makes the finite symmetries transparent. The mixed chart is where we test logarithmic regularity, the negative twist, and the overlaps between the two boundary components.

Let

$$(8.6) \quad U_0 = \{ZAB \neq 0\}.$$

On U_0 , put

$$x = \frac{X}{Z}, \quad y = \frac{Y}{Z},$$

and

$$(8.7) \quad a = x^3 + 2y^3 + 1, \quad b = 2x^3 + y^3 + 1.$$

The ordinary invariant two-jet frame on U_0 is

$$(8.8) \quad x', \quad y', \quad W_{xy} := x'y'' - x''y'.$$

The mixed logarithmic chart is taken in $\{X \neq 0\}$. Put

$$u = \frac{Y}{X}, \quad z = \frac{Z}{X}.$$

Then

$$(8.9) \quad a_X = 1 + 2u^3 + z^3, \quad b_X = 2 + u^3 + z^3.$$

Set

$$(8.10) \quad q = \left(\log \frac{a_X}{b_X} \right)'.$$

A direct computation gives

$$(8.11) \quad q = \frac{3u^2(3+z^3)}{a_X b_X} u' + \frac{3z^2(1-u^3)}{a_X b_X} z'.$$

The determinant of the frame (z', q) is therefore

$$(8.12) \quad \Delta_X := a_{X,u} b_X - a_X b_{X,u} = 3u^2(3+z^3).$$

We define

$$(8.13) \quad U_X = \{X \neq 0, u(3+z^3) \neq 0\}.$$

On this chart, the logarithmic invariant two-jet frame is

$$(8.14) \quad z', \quad q, \quad W_X := qz'' - q'z'.$$

The two sets U_0 and U_X cover the complement of a finite set. Indeed,

$$\mathbb{P}^2 \setminus (U_0 \cup U_X) \subset \{ZAB = 0\} \cap \{XY(3X^3 + Z^3) = 0\},$$

and the right-hand side is finite because none of the curves $Z = 0$, $A = 0$, or $B = 0$ is a component of $XY(3X^3 + Z^3) = 0$. Since the jet differential sheaves used here are locally free on the normal crossing locus, it is enough to test regularity on $U_0 \cup U_X$.

For the unequal-degree rows of (8.1), the same construction uses the projectively invariant logarithmic form. Put

$$g = \gcd(d_1, d_2), \quad \alpha = \frac{d_2}{g}, \quad \beta = \frac{d_1}{g},$$

and on the X -chart define

$$(8.15) \quad q = \alpha d \log a_X - \beta d \log b_X, \quad \Delta = \alpha a_{X,u} b_X - \beta a_X b_{X,u}, \quad E = \alpha a_{X,z} b_X - \beta a_X b_{X,z}.$$

Then

$$(8.16) \quad q = \frac{\Delta u' + E z'}{a_X b_X}, \quad u' = \frac{a_X b_X}{\Delta} q - \frac{E}{\Delta} z'.$$

The frame generator may replace q by any nonzero rational scalar multiple without changing the kernel problem over $\mathbb{Q}$. The implementation uses this freedom to keep the mixed-chart numerators integral. For the diagonal nonlinear models it uses

$$q_{\text{cmp}} = \frac{1}{d_1} d \log a_X - \frac{1}{d_2} d \log b_X = \frac{g}{d_1 d_2} q.$$

For the mixed conic model $a_X = u + z^2$, of bidegree $(2, e)$, it instead uses

$$q_{\text{cmp}} = d \log a_X - \frac{2}{e} d \log b_X = \frac{g}{e} q, \quad g = \gcd(2, e).$$

Thus $q_{\text{cmp}} = q/5$ for $e = 5$, whereas $q_{\text{cmp}} = q/3$ for $e = 6$. This is the normalization whose degree factors cancel in the corresponding mixed numerators. For the two line models the program uses the integral form q displayed below. Thus in every model the resulting Δ_{cmp} , E_{cmp} , and every matrix entry produced by the program are integers. Each rescaling is an invertible change of frame over $\mathbb{Q}$; it is not being asserted to be invertible after reduction modulo every prime. Instead, the program first constructs its own integral matrix in this normalized frame. Full column rank of that integral matrix modulo one prime implies full column rank over $\mathbb{Q}$, and the rational change of frame then identifies its kernel with the kernel of the invariant form used in (8.15). In particular, the proof-effective prime 5 for the conic–quintic case is legitimate even though 5 divides the denominator in the rational rescaling $q_{\text{cmp}} = q/5$. The order is important: the change of frame is made over characteristic zero, the normalized matrix is then reconstructed with integral entries, and only this integral matrix is reduced modulo 5. No occurrence of the rational scalar $1/5$ is reduced in $\mathbb{F}_5$.

Likewise, the program's cyclic blocks are defined directly by congruences of integer monomial exponents and, where present, by the explicit exchange involution. Their construction does not invoke semisimplicity of a finite-group representation over the proof field. Semisimplicity is used only in characteristic zero to show that these blocks exhaust the candidate space. Hence a proof prime may divide the order of the displayed symmetry group without invalidating the integral block computation. For the two line models these formulas are especially simple. When $e = 8$,

$$(8.17) \quad \begin{aligned} a_X &= u, & b_X &= 1 + u^8 + z^8 + u^4, & q &= 8 d \log a_X - d \log b_X, \\ \Delta &= 8 + 8z^8 + 4u^4, & E &= -8uz^7. \end{aligned}$$

When $e = 9$,

$$(8.18) \quad \begin{aligned} a_X &= u, & b_X &= 1 + u^9 + z^9 + u^3 z^3, & q &= 9 d \log a_X - d \log b_X, \\ \Delta &= 9 + 9z^9 + 6u^3 z^3, & E &= -9uz^8 - 3u^4 z^2. \end{aligned}$$

In both cases z, u, b_X, Δ are pairwise coprime. For $e = 8$, put $\delta_0 = \Delta/4$. The identity

$$2b_X - \delta_0 = u^4(2u^4 + 1)$$

reduces the only nonobvious gcd to u and the factors of $2u^4 + 1$; the former does not divide b_X , and no nonconstant polynomial in u divides the polynomial b_X , which is monic in z . For $e = 9$, a common factor of b_X and $\Delta = 9b_X - u(b_X)_u$ would, after excluding u , divide $3u^6 + z^3$. On each component $z = \lambda u^2$, $\lambda^3 = -3$, the restriction of b_X is

$$1 - 2u^9 - 27u^{18},$$

which is not identically zero. This proves the asserted coprimality.

The coefficients α, β are exactly what make q independent of the projective trivialization. For every test pair in (8.1), none of Z, A, B is a component of the homogeneous polynomial $X\Delta$. Hence

$$U_0 = \{ZAB \neq 0\}, \quad U_X = \{X\Delta \neq 0\}$$

again cover the complement of a finite set. The general logarithmic frame is (z', q, W_X) with $W_X = qz'' - q'z'$. Formulas (8.10)–(8.12) are the specialization $d_1 = d_2 = 3$.

8.4. An auxiliary logarithmic compactification and a finite ambient normal form.

The ordinary frame on U_0 is convenient for symmetry, but it is not adapted to the line at infinity. In particular, the negative twist cannot be encoded by simply subtracting t from the degree of each numerator in the ordinary frame: cancellations between different ordinary basis terms may remove the apparent poles at infinity. We therefore separate the finite ambient bound from the twisting condition.

Let

$$L := \{Z = 0\}, \quad D^+ := D + L.$$

For every test pair in (8.1), the divisor D^+ is simple normal crossing. Multiplication by the section $Z^t \in H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(t))$ gives an injection

$$(8.19) \quad E_{2,m}T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t) \xrightarrow{\cdot Z^t} E_{2,m}T_{\mathbb{P}^2}^*(\log D) \hookrightarrow E_{2,m}T_{\mathbb{P}^2}^*(\log D^+).$$

The image of the first arrow consists of logarithmic two-jet differentials for D that vanish to order at least t along L . Since $Z = 1$ on the affine chart used to define U_0 , multiplication by Z^t does not change the local expression on U_0 . Thus we may first generate a finite ambient space inside the larger logarithmic bundle for D^+ and impose the removal of the extra logarithmic pole along L , together with the order- t vanishing, on U_X .

We begin with the local pole estimate used for all three components of D^+ .

Lemma 8.1 (Layerwise logarithmic pole bound). *Let $s = 0$ be a smooth local branch of a logarithmic divisor on a surface, and let v be a transverse coordinate. Put*

$$\lambda = \frac{s'}{s}, \quad W_{sv} = s'v'' - s''v'.$$

Then the logarithmic Wronskian satisfies the triangular identity

$$(8.20) \quad \lambda v'' - \lambda' v' = \frac{1}{s} W_{sv} + \frac{(s')^2 v'}{s^2}.$$

Consequently, in weighted degree m , the coefficient of the ordinary layer W_{sv}^k has pole order at most

$$n_k := m - 2k$$

along $s = 0$. More precisely, the coefficient of

$$(s')^p (v')^{m-3k-p} W_{sv}^k$$

has pole order at most $p + k$.

Proof. Consider a logarithmic monomial

$$\lambda^a (v')^b (\lambda v'' - \lambda' v')^r, \quad a + b = m - 3r.$$

A contribution to the ordinary Wronskian layer W_{sv}^k can occur only when $r \geq k$. Choose the leading term $s^{-1}W_{sv}$ in exactly k of the r Wronskian factors and the lower term $s^{-2}(s')^2 v'$ in the remaining $r - k$ factors. The pole order is then

$$a + k + 2(r - k).$$

The resulting number of factors s' outside W_{sv}^k is

$$p = a + 2(r - k),$$

so the pole order equals $p + k$. Since $p \leq m - 3k$, this is at most $m - 2k$. The statement follows by linearity. $\square$

Write

$$K = \lfloor m/3 \rfloor, \quad n_k = m - 2k, \quad s_k = m - 3k, \quad d_k^+ = (s - 1)m - (2s - 1)k.$$

Proposition 8.2 (Finite ambient normal form on U_0). *Every global section of*

$$E_{2,m}T_{\mathbb{P}^2}^*(\log D^+)$$

restricts to U_0 in the form

$$(8.21) \quad \omega_0 = \sum_{k=0}^K \sum_{i+j=s_k} \frac{P_{i,j,k}(x,y)}{a^{n_k}b^{n_k}} (x')^i (y')^j W_{xy}^k,$$

where

$$(8.22) \quad P_{i,j,k} \in \mathbb{C}[x,y], \quad \deg P_{i,j,k} \leq d_k^+ = (s-1)m - (2s-1)k.$$

The exponent $n_k = m - 2k$ and the degree bound d_k^+ are necessary ambient bounds. Arbitrary polynomials satisfying (8.22) need not define a global section; the missing compatibility conditions are imposed on U_X in the next subsection.

Proof. Apply Lemma 8.1 to the branches $a = 0$ and $b = 0$. The coefficient of W_{xy}^k has pole order at most n_k along each branch. After multiplication by $a^{n_k}b^{n_k}$, every coefficient is regular on the affine plane, hence is a polynomial. This proves the denominator assertion.

It remains to control the degree by using the extra logarithmic component L . On the X -chart write

$$z = \frac{Z}{X}, \quad u = \frac{Y}{X}, \quad x = \frac{1}{z}, \quad y = \frac{u}{z}.$$

The ordinary jets satisfy

$$(8.23) \quad x' = -\frac{z'}{z^2}, \quad y' = \frac{zu' - uz'}{z^2}, \quad W_{xy} = -\frac{W_{zu}}{z^3},$$

where $W_{zu} = z'u'' - z''u'$. Moreover,

$$a = \frac{a_X}{z^{d_1}}, \quad b = \frac{b_X}{z^{d_2}}.$$

Thus the k -th layer of (8.21) becomes

$$(8.24) \quad \frac{z^{(s-2)m-(2s-3)k}}{a_X^{n_k}b_X^{n_k}} \sum_{i+j=s_k} (-1)^{i+k} P_{i,j,k}(z^{-1}, uz^{-1}) (z')^i (zu' - uz')^j W_{zu}^k.$$

For $0 \leq \ell \leq s_k$, the coefficient of

$$(z')^{s_k-\ell} (u')^\ell W_{zu}^k$$

is a triangular linear combination of the polynomials $P_{s_k-j,j,k}(z^{-1}, uz^{-1})$ with $j \geq \ell$; its diagonal term is a nonzero scalar multiple of

$$z^{(s-2)m-(2s-3)k+\ell} P_{s_k-\ell,\ell,k}(z^{-1}, uz^{-1}).$$

By the refined part of Lemma 8.1, a section logarithmic along $L = \{z = 0\}$ may have pole order at most

$$(s_k - \ell) + k$$

in this coefficient. Starting with $\ell = s_k$ and descending in ℓ , the triangularity shows that no polynomial $P_{s_k-\ell,\ell,k}$ can have degree exceeding

$$(s-2)m - (2s-3)k + \ell + (s_k - \ell) + k = (s-1)m - (2s-1)k.$$

This proves (8.22). $\square$

The important point is that the negative twist t does not appear in (8.22). The degree bound describes a finite ambient space with an allowed logarithmic pole along L . The twist is imposed by requiring the transformed coefficients to lose that pole and to vanish to order t along L .

8.5. **Regularity and twisting test on U_X .** On $U_0 \cap U_X$ one has

$$(8.25) \quad x = \frac{1}{z}, \quad y = \frac{u}{z}, \quad a = \frac{a_X}{z^{d_1}}, \quad b = \frac{b_X}{z^{d_2}}.$$

The first derivatives transform as

$$(8.26) \quad x' = -\frac{z'}{z^2}, \quad y' = \frac{u'z - uz'}{z^2}.$$

Set

$$H := a_X b_X, \quad E := \alpha_{a_X, z} b_X - \beta_{a_X} b_{X, z}, \quad \Delta_X := \alpha_{a_X, u} b_X - \beta_{a_X} b_{X, u}.$$

Equation (8.11) gives

$$(8.27) \quad u' = \frac{H}{\Delta_X} q - \frac{E}{\Delta_X} z'.$$

The pair (z', q) is a logarithmic one-jet frame for D on U_X , and

$$W_X = qz'' - q'z'$$

is the associated invariant Wronskian. Substituting (8.25), (8.26), and (8.27) into (8.21) and collecting in the mixed frame gives

$$(8.28) \quad \omega_0|_{U_X} = \sum_{k=0}^K \sum_{j=0}^{s_k} R_{j,k}(u, z) (z')^{s_k-j} q^j W_X^k.$$

Proposition 8.3 (Exact extension and twisting criterion). *A candidate (8.21) is the restriction of a section*

$$\omega \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t))$$

if and only if every coefficient in (8.28) satisfies

$$(8.29) \quad R_{j,k} \in z^t \mathbb{C}[u, z, \Delta_X^{-1}].$$

Equivalently, after clearing a fixed bounded denominator, the condition is a finite collection of polynomial divisibility equations that are homogeneous and linear in the coefficients of the $P_{i,j,k}$.

Proof. The mixed frame (z', q, W_X) is regular for the logarithmic jet bundle associated with D on U_X . Hence a section of the untwisted bundle is regular on U_X exactly when its scalar coefficients lie in

$$\mathcal{O}(U_X) = \mathbb{C}[u, z, \Delta_X^{-1}].$$

Under the injection (8.19), the image of a section twisted by $\mathcal{O}(-t)$ vanishes to order at least t along $L = \{z = 0\}$. This is precisely condition (8.29).

Since $U_0 \cup U_X$ misses only finitely many points and the jet bundle is locally free, a section satisfying this condition extends uniquely across the missing codimension-two set. This proves the equivalence. $\square$

For computation, the denominator clearing can be made uniform and explicit. Write

$$u' = Aq + Bz', \quad A = H/\Delta_X, \quad B = -E/\Delta_X.$$

Differentiating with respect to the source variable and substituting $u' = Aq + Bz'$ gives the exact identity

$$(8.30) \quad u'' = Aq' + Bz'' + AA_u q^2$$

$$(8.31) \quad + (A_u B + A_z + AB_u) qz' + (BB_u + B_z)(z')^2.$$

Consequently,

$$(8.32) \quad \begin{aligned} W_{zu} = & -AW_X + AA_u q^2 z' \\ & + (A_u B + A_z + AB_u) q(z')^2 + (BB_u + B_z)(z')^3. \end{aligned}$$

This is an identity in the localized differential-polynomial ring. Since A, B have one power of Δ_X in the denominator, their first partial derivatives have at most two, and every coefficient in (8.32) has Δ_X -denominator order at most three. A source layer W_{zu}^r can contribute to the target layer W_X^k only when $r \geq k$. The $m - 3r$ first-jet factors contribute at most one power of Δ_X^{-1} each, the k leading Wronskian factors contribute one power each, and the remaining $r - k$ lower Wronskian factors contribute at most three powers each. Therefore the total exponent of Δ_X is bounded by

$$(8.33) \quad (m - 3r) + k + 3(r - k) = m - 2k = n_k.$$

No substitution introduces a_X or b_X in a denominator beyond the original source exponent, and $n_r \leq n_k$ for $r \geq k$.

We also record the z -bound explicitly. Before replacing u' and W_{zu} by the mixed frame, a monomial from source layer r with a numerator of degree d carries the factor

$$z^{sn_r - 2(m - 3r) - 3r - d} = z^{(s - 2)m - (2s - 3)r - d}.$$

Since $d \leq (s - 1)m - (2s - 1)r$, this exponent is at least

$$(s - 2)m - (2s - 3)r - ((s - 1)m - (2s - 1)r) = -(m - 2r) = -n_r.$$

The substitutions (8.27) and (8.32) introduce no additional powers of z^{-1} . Hence the pole order in z is at most $n_r \leq n_k$. Consequently,

$$(8.34) \quad \mathcal{N}_{j,k} := z^{n_k} a_X^{n_k} b_X^{n_k} \Delta_X^{n_k} R_{j,k} \in \mathbb{C}[u, z].$$

The same valuation argument applies to every row of (8.1), including the unequal-degree rows. Indeed, $A + B + L$ is SNC, so z, a_X, b_X are pairwise coprime, and the check following (8.16) says precisely that

$$\gcd(z a_X b_X, \Delta_X) = 1.$$

Thus localizing at Δ_X changes none of the valuations along the prime factors of $z a_X b_X$. This is the required bridge from the general coprimality $\gcd(A, B) = 1$ to the concrete divisibility test.

For example, for the special pair (8.2), the four factors

$$z, \quad a_X, \quad b_X, \quad \Delta_X$$

are pairwise coprime in $\mathbb{C}[u, z]$. Indeed, z divides none of the other three. If an irreducible factor divided both a_X and b_X , it would divide $a_X - b_X = u^3 - 1$; after setting $u^3 = 1$, however, both polynomials become $3 + z^3$, so no factor of $u^3 - 1$ divides them identically. Moreover, u divides neither a_X nor b_X . Finally, for every root ζ of $\zeta^3 = -3$ one has

$$a_X(u, \zeta) = 2(u^3 - 1), \quad b_X(u, \zeta) = u^3 - 1,$$

which are not the zero polynomial in u ; hence no irreducible factor of $3 + z^3$ divides a_X or b_X . Since $\mathbb{C}[u, z]$ is a unique factorization domain, this verifies the general coprimality statement directly in the cubic case. The valuation conditions along $z = 0$, $a_X = 0$, and $b_X = 0$ can therefore be combined for every test pair. Hence (8.29) is equivalent to the concrete divisibility condition

$$(8.35) \quad z^{n_k + t} a_X^{n_k} b_X^{n_k} \mid \mathcal{N}_{j,k} \quad \text{in } \mathbb{C}[u, z].$$

For the conic and cubic models, the fixed product in (8.35) may be handled by the corresponding exact polynomial-remainder test. For a line pair, however, $a_X = u$, and treating the whole product as a univariate polynomial in z would give a nonconstant leading coefficient. We instead use pairwise coprimality and impose the following three equivalent conditions on every cleared numerator $\mathcal{N}_{j,k}$:

- (i) no monomial has z -degree below $n_k + t$;
- (ii) no monomial has u -degree below n_k ;
- (iii) the remainder modulo the monic polynomial $b_X^{n_k}$ in $K[u][z]$ is zero, where $K = \mathbb{C}$ for the characteristic-zero system and $K = \mathbb{F}_p$ for a certificate matrix.

The three row families are kept disjoint in the certificate. Hence all divisibility conditions are imposed simultaneously, and each is homogeneous and linear in the coefficients of the $P_{i,j,k}$. The triangularity in the Wronskian degree permits the equations to be generated from the largest k downward.

8.6. The involution and character reductions. The involution $\sigma(X, Y, Z) = (Y, X, Z)$ preserves U_0 and acts on the ordinary frame by

$$x \leftrightarrow y, \quad x' \leftrightarrow y', \quad W_{xy} \mapsto -W_{xy}.$$

If one fixes an eigenvalue $\varepsilon = \pm 1$ and imposes

$$(8.36) \quad \sigma^* \omega_0 = \varepsilon \omega_0,$$

then the coefficients in (8.21) satisfy

$$(8.37) \quad P_{i,j,k}(x, y) = \varepsilon(-1)^k P_{j,i,k}(y, x).$$

Thus only one representative from each orbit

$$(i, j, x^r y^s) \longleftrightarrow (j, i, x^s y^r)$$

needs to be generated. If an orbit is fixed and $\varepsilon(-1)^k = -1$, its coefficient is forced to be zero.

The diagonal $(\mu_3)^2$ -action gives a further decomposition. For

$$[X : Y : Z] \mapsto [\alpha X : \beta Y : Z], \quad \alpha^3 = \beta^3 = 1,$$

one has

$$x \mapsto \alpha x, \quad y \mapsto \beta y, \quad x' \mapsto \alpha x', \quad y' \mapsto \beta y', \quad W_{xy} \mapsto \alpha \beta W_{xy}.$$

Therefore the monomial

$$x^r y^s (x')^i (y')^j W_{xy}^k$$

has character

$$(8.38) \quad (r + i + k, s + j + k) \pmod{3}.$$

For a fixed character $(\chi_1, \chi_2) \in (\mathbb{Z}/3\mathbb{Z})^2$, we generate only the monomials satisfying

$$(8.39) \quad r + i + k \equiv \chi_1 \pmod{3}, \quad s + j + k \equiv \chi_2 \pmod{3}.$$

The involution σ exchanges the character (χ_1, χ_2) with (χ_2, χ_1) . Hence the nine characters split into six orbits:

$$(8.40) \quad (0, 0), (1, 1), (2, 2), (0, 1) \leftrightarrow (1, 0), (0, 2) \leftrightarrow (2, 0), (1, 2) \leftrightarrow (2, 1).$$

For diagonal characters one imposes σ -eigenconditions inside the character space. For an off-diagonal orbit, an eigenvector is determined by its component in either one of the two exchanged character spaces.

Lemma 8.4 (The cubic symmetry blocks exhaust the candidate space). *Let V be any finite-dimensional candidate space in the above normal form, and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with every one of the twelve character-eigenvalue blocks indexed by (8.40) and $\varepsilon \in \{+1, -1\}$ is zero.*

Proof. Over $\mathbb{C}$, the finite abelian group $(\mu_3)^2$ acts semisimply, so $V = \bigoplus_{\chi} V_{\chi}$. The equations are equivariant and hence their kernel has the same decomposition. The involution sends V_{χ} to $V_{\sigma\chi}$. If $\chi = \sigma\chi$, the involution is diagonalizable with eigenvalues ± 1 . If $\chi \neq \sigma\chi$ and $0 \neq v \in K \cap V_{\chi}$, then at least one of $v + \sigma v$ and $v - \sigma v$ is nonzero; these vectors lie respectively in the $+1$ and -1 eigenspaces of $V_{\chi} \oplus V_{\sigma\chi}$. Thus every nonzero kernel contains a nonzero vector in one of the listed blocks. The converse is immediate. $\square$

Lemma 8.5 (The conic-pair symmetry blocks exhaust the candidate space). *For either model (2, e), let V be the finite candidate space and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with every character-eigenvalue block indexed by an orbit of $c \mapsto -c$ in $\mathbb{Z}/e\mathbb{Z}$ and by $\varepsilon \in \{+1, -1\}$ is zero.*

Proof. Over $\mathbb{C}$, the cyclic group μ_e acts semisimply, and equivariance of the overlap and divisibility equations gives $K = \bigoplus_c (K \cap V_c)$. Coordinate exchange sends V_c to V_{-c} and preserves K . On a fixed character ($c = -c$), it is diagonalizable with eigenvalues ± 1 . On a two-element orbit $\{c, -c\}$, every nonzero $v \in K \cap V_c$ yields a nonzero vector among $v + \sigma v$ and $v - \sigma v$ in the corresponding $+1$ or -1 block. Thus a nonzero kernel meets one of the enumerated blocks, and the converse is immediate. $\square$

Lemma 8.6 (The line-pair character blocks exhaust the candidate space). *For either line model $(1, e)$, let V be the finite candidate space and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with each of the e character spaces defined by (8.5) is zero.*

Proof. The cyclic group μ_e acts semisimply in characteristic zero, and all overlap and divisibility equations are equivariant. Both V and K are therefore direct sums of their e character spaces. $\square$

Proposition 8.7 (Finite matrix equivalence). *Fix one of the test pairs in (8.1) and integers $m, t \geq 1$. Let V_m be the finite coefficient space in (8.21)–(8.22), and let*

$$\Phi_{m,t}: V_m \longrightarrow W_{m,t}$$

record the coefficients of all forbidden remainders in (8.35); for a line pair, $\Phi_{m,t}$ records the three row families listed after that equation. Restriction to U_0 induces a natural isomorphism

$$H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)) \simeq \ker \Phi_{m,t}.$$

Moreover, the symmetry decompositions above identify this kernel with the direct sum of the kernels of the corresponding block matrices.

Proof. Proposition 8.2 sends every global section to a unique vector of V_m . Proposition 8.3, together with the coprimality and denominator bounds proved above, says that such a vector comes from a twisted global section exactly when all the remainders recorded by $\Phi_{m,t}$ vanish. The resulting section on $U_0 \cup U_X$ extends uniquely across the finite complement. This proves the first assertion. Equivariance of the equations and Lemmas 8.4, 8.5, and 8.6 prove the second. $\square$

The implementation also checks equivariance after the full chart transformation. If a cleared target row contains the monomial $u^U z^Z$, has q -degree Q , and lies in target Wronskian layer k , its character in the source convention is

$$(8.41) \quad -2U - Z + 2(m - 2k) + Q \pmod{e}.$$

For every complete transformed column in the $m = 3$ self-test, each nonzero target row has the same character as the source column. This detects, in particular, omission of the denominator shift in (8.5).

8.7. Counting the ambient variables. Let

$$M(d) = \binom{d+2}{2}$$

be the number of monomials in two variables of total degree at most d . Since the negative twist is imposed by the equations on U_X , rather than by deleting monomials on U_0 , the ambient variable count depends on m but not on t :

$$(8.42) \quad N_{\text{raw}}^+(m; s) = \sum_{k=0}^{\lfloor m/3 \rfloor} (m - 3k + 1) \binom{(s-1)m - (2s-1)k + 2}{2}.$$

For a line pair $(1, e)$, so that $s = e + 1$, this specializes to

$$(8.43) \quad N_e(m) = \sum_{0 \leq k \leq \lfloor m/3 \rfloor} (m - 3k + 1) \binom{em - (2e+1)k + 2}{2}.$$

The exact counts for the eight line-pair cases are

$$(8.44) \quad \begin{array}{c|c|c|c|c} (d_1, d_2) & (m, t) & N_e(m) & \text{blocks} & \text{largest block} \\ \hline (1, 8) & (3, 2) & 1,336 & 8 & 174 \\ (1, 8) & (4, 2) & 3,077 & 8 & 398 \\ (1, 8) & (5, 2) & 6,066 & 8 & 779 \\ (1, 8) & (6, 3) & 10,807 & 8 & 1,382 \\ (1, 8) & (7, 3) & 17,876 & 8 & 2,279 \\ (1, 8) & (8, 3) & 27,849 & 8 & 3,543 \\ (1, 9) & (3, 3) & 1,669 & 9 & 186 \\ (1, 9) & (4, 3) & 3,857 & 9 & 429 \end{array}$$

Thus the line-octic calculation has 48 blocks and 67,011 ambient columns in total; the line-nonic calculation has 18 blocks and 5,526 columns.

The remainder of this subsection specializes to $(d_1, d_2) = (3, 3)$, so $s = 6$ and the degree bound becomes $5m - 11k$.

For a fixed character (χ_1, χ_2) , define

$$(8.45) \quad M_d(a, b) := \#\{(r, s) \in \mathbb{Z}_{\geq 0}^2 : r + s \leq d, r \equiv a \pmod{3}, s \equiv b \pmod{3}\}.$$

Then

$$(8.46) \quad N_{\chi_1, \chi_2}^+(m) = \sum_{k=0}^{\lfloor m/3 \rfloor} \sum_{i+j=m-3k} M_{5m-11k}(\chi_1 - i - k, \chi_2 - j - k),$$

where all congruences are understood modulo 3. For an off-diagonal character orbit, the number of variables in a σ -eigensystem is the number in either exchanged character. For a diagonal character (χ, χ) , the σ -even and σ -odd dimensions are

$$(8.47) \quad N_{\chi}^{\pm,+}(m) = \frac{1}{2} (N_{\chi, \chi}^+(m) \pm T_{\chi}^+(m)),$$

where

$$(8.48) \quad T_{\chi}^+(m) = \sum_{\substack{0 \leq k \leq \lfloor m/3 \rfloor \\ m-3k \text{ even}}} (-1)^k \# \left\{ r \geq 0 : 2r \leq 5m - 11k, r + \frac{m-3k}{2} + k \equiv \chi \pmod{3} \right\}.$$

For each twist t , the following table records the endpoint $m = 5t - 1$ of the certified cubic range. The last column is the largest block after both the character and involution decompositions.

$$(8.49) \quad \begin{array}{c|c|c|c} t & m & N_{\text{raw}}^+ & \max N_{\text{sym}}^+ \\ \hline 1 & 4 & 1,265 & 156 \\ 2 & 9 & 16,511 & 1,924 \\ 3 & 14 & 76,905 & 8,830 \end{array}$$

Thus the largest ambient candidate space occurs at $(m, t) = (14, 3)$ and has 76,905 variables. After the $(\mu_3)^2$ -character decomposition and the involution, the largest block has 8,830 variables. The raw ambient total over all pairs

$$1 \leq t \leq 3, \quad 1 \leq m \leq 5t - 1$$

is 280,798. These variables are never placed in one matrix: the computation splits by (m, t) , character orbit, and involution eigenvalue. Within each such block, the Wronskian layers organize the downward generation of equations, but the rank test is performed on the full coupled matrix containing all layers. Although the ambient variables do not depend on t , the divisibility equations (8.35) become stronger as t increases.

8.8. Exact modular rank verification. For each degree pair and each (m, t) in its finite range, the computation is as follows.

- (1) Use the injection (8.19) and generate the finite ambient normal form (8.21) on U_0 , with denominator exponent $m - 2k$ and degree bound $(s - 1)m - (2s - 1)k$. Do not subtract t from this degree bound.
- (2) For $(3, 3)$, split the ambient variables according to the $(\mu_3)^2$ -characters and impose the involution eigencondition. For $(2, e)$ use the cyclic character and involution blocks described above. For $(1, e)$ use exactly the e shifted cyclic characters in (8.5), without an involution. Retain one full unsplit sparse basis for $(3, 4)$.
- (3) Transform the candidate to U_X using (8.25), (8.26), (8.27), and the triangular Wronskian formula (8.32).
- (4) Collect the expression in the regular logarithmic frame (z', q, W_X) and form the cleared polynomials $\mathcal{N}_{j,k}$ in (8.34).
- (5) Impose the exact divisibility conditions (8.35). For a line pair use the three separate row families listed after that equation. These conditions simultaneously remove the spurious logarithmic pole along L , impose the twist $\mathcal{O}(-t)$, and cancel all forbidden poles along A and B .
- (6) Choose a prime for which the adopted normalized integral matrix is defined, reduce that integral matrix to the corresponding finite field, and verify full column rank. If the frame was obtained by a rational rescaling such as $q_{\text{cmp}} = q/5$, perform that rescaling over characteristic zero and reconstruct the integral matrix before reduction; do not reduce the rational change-of-frame scalar itself. Repeat over a second prime as an implementation check; only one proof-effective prime is logically necessary.

For $(3, 3)$ it suffices to check the twelve genuine second-level pairs

$$(8.50) \quad 3 \leq m \leq 14, \quad t = t_0(m) := \left\lceil \frac{m+1}{5} \right\rceil.$$

Indeed, if $t \geq t_0(m)$ and a section with twist $-t$ existed, multiplication by a nonzero form of degree $t - t_0(m)$ would inject it into the space with twist $-t_0(m)$. The implementation also checks $m = 1, 2$ as low-degree self-tests. The remaining minimal pairs are exactly those displayed after Lemma 1.2.

The passage from modular rank to characteristic zero is elementary but decisive. Every cleared matrix has integral entries. Full column rank modulo a prime p exhibits a maximal minor that is nonzero modulo p ; that minor is a nonzero integer and remains nonzero over $\mathbb{Q}$.

The certificate program constructs the matrices from (8.35). For every cubic pair in (8.50), all twelve character–eigenvalue matrices have full column rank. Lemma 8.4 then gives zero kernel for the entire cubic candidate space. For $(2, e)$, every cyclic character–eigenvalue block has full column rank; the $(3, 4)$ unsplit matrix does as well. For $(1, e)$, every shifted cyclic character block has full column rank; Lemma 8.6 therefore gives zero kernel. The proof primes are recorded in Table 1. For each block, the program writes the degree pair, (m, t) , row and column numbers, rank, nullity, nonzero pivot product and pivot hash, modulus, and elapsed time, both as plain text and in JSON Lines format. The complete source, machine-readable certificates, checksums, and reproduction instructions are included in the supplementary archive cited above. A separate verifier audits the expected case and block coverage and the internal consistency of every certificate record; the frozen source and binaries permit the rank computations themselves to be reproduced.

The proof moduli displayed in the table already suffice. The archive also contains reruns at other primes to test the implementation independently; those reruns are reproducibility checks, not additional mathematical hypotheses. For $(3, 3)$, the newly required case $(m, t) = (9, 2)$ was replayed in full at prime 7. For $(2, 5)$, the additional case $(m, t) = (8, 3)$ has 15,849 columns distributed among six character–eigenvalue blocks. All six blocks have full column rank modulo 5, and an independent replay gives full column rank modulo 7 as well.

TABLE 1. Finite-rank certificate summary. “Columns” is summed over the indicated independent blocks; in every row the summed rank equals the summed number of columns, so the kernel dimension is zero. The proof modulus is the prime used for the characteristic-zero inference. For $(3, 3)$, the totals include the 12 theorem-relevant pairs in (8.50) and the two low-weight implementation self-tests $m = 1, 2$.

pair	cases	blocks	columns	rank	kernel dim.	proof modulus
$(3, 3)$	14	168	280,798	280,798	0	5
$(3, 4)$	1	1	781	781	0	5
$(2, 6)$	1	8	1,040	1,040	0	7
$(2, 5)$	6	36	38,286	38,286	0	5
$(1, 9)$	2	18	5,526	5,526	0	5
$(1, 8)$	6	48	67,011	67,011	0	5

Prime 5 is a bad reduction for one $(2, 6)$ block: there the rank is $92/93$. It will be retained in the archive only as a diagnostic and is not used in the proof. The proof-effective $(2, 6)$ certificate is the full-rank prime-7 run, independently reproduced at prime 11.

For both line pairs, prime 5 supplies the proof-effective full-pivot certificates and prime 11 supplies an independent full-rank replay. Prime 7 is excluded: some relevant integral minors vanish modulo 7, and the degree-nine test curve has bad reduction there because $1^3 = -27$ in $\mathbb{F}_7$. None of the prime-7 diagnostics is used in the proof.

By the maximal-minor argument above, the proof-effective ranks in Table 1 give zero kernel over $\mathbb{Q}$, and hence over $\mathbb{C}$. This proves the required vanishing for every test pair in (8.1).

To pass to general pairs of the fixed degrees, let $\mathcal{U}$ be the Zariski-open parameter space of ordered smooth transverse pairs, let $\mathcal{D} \subset \mathbb{P}^2 \times \mathcal{U}$ be the relative simple-normal-crossing divisor, and let $\mathcal{E}_{m,t}$ be the relative logarithmic invariant two-jet bundle twisted by $\mathcal{O}_{\mathbb{P}^2}(-t)$. Properness of the projection $\mathbb{P}^2 \times \mathcal{U} \rightarrow \mathcal{U}$ and upper semicontinuity of fiberwise cohomology imply that

$$(C_1, C_2) \mapsto h^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log(C_1 + C_2)) \otimes \mathcal{O}(-t))$$

is upper semicontinuous. Vanishing at the special pair therefore holds on a nonempty Zariski-open subset of $\mathcal{U}$. There are only finitely many pairs (m, t) under consideration; intersecting their nonempty open loci proves Lemma 1.2.

9. THE COMPUTATIONAL FRONTIER FOR THE SMT CONSTANTS

The constants 57, 45, and 69 are convenient explicit choices, not claimed to be optimal. Their role is to show that the two-component reduction and the finite algorithm already give an effective Second Main Theorem on an ordinary computer. Once the geometric argument is fixed, enlarging the certified finite-vanishing range improves the constants without requiring a new geometric idea.

9.1. The limiting constants of the present method. The smallest integer constants permitted by the present two-branch organization are conditionally 27, 17, and 18 for $(d_1, d_2) = (3, 3), (2, 5), (1, 8)$, respectively. The corresponding ratio splits and zero-locus endpoints are

(d_1, d_2)	ρ_*	$3/\tau_1(\rho_*)$
$(3, 3)$	$1/7$	$(27 + \sqrt{701})/2 < 27,$
$(2, 5)$	$2/9$	$(207 + 54\sqrt{14})/25 < 17,$
$(1, 8)$	$4/19$	$(222 + \sqrt{47859})/25 < 18.$

Relative to Lemma 1.2, these improvements require exactly the following additional minimal vanishings:

$$\begin{array}{l|l}
 (3, 3) & (5, 1), (6, 1), (10, 2), (11, 2), (12, 2), (13, 2), (15, 3), \\
 & (16, 3), (17, 3), (18, 3), (19, 3), (20, 3), \\
 (2, 5) & (3, 1), (4, 1), (6, 2), (7, 2), (8, 2), (9, 3), \\
 & (10, 3), (11, 3), (12, 3), (13, 3), \\
 (1, 8) & (3, 1), (4, 1), (6, 2), (7, 2), (8, 2), (9, 2), (10, 3), \\
 & (11, 3), (12, 3), (13, 3), (14, 3).
 \end{array}$$

Indeed, these are precisely the integral pairs with $1 \leq t \leq 3$ above the indicated split, after twist monotonicity removes all nonminimal twists. Thus this list, rather than a change in the analytic or geometric argument, is the remaining obstruction to the limiting constants.

9.2. Computational scale and feasibility. All the additional targets are governed by the same finite map $\Phi_{m,t}$ and the same symmetry blocks as in Section 8. In particular, no new symbolic normal form is required: only the weight range is enlarged. The following counts are produced by the frozen program’s matrix-free `--estimate-only` path, using exactly the basis and symmetry conventions of the certificate computation:

(d_1, d_2)	new targets	$\sum N_{\text{raw}}$	shards	N_{raw} at the endpoint	B_{max}
(3, 3)	12	1,248,727	144	280,686	31,931
(2, 5)	10	288,129	60	84,865	16,973
(1, 8)	11	704,491	88	195,750	24,725
total	33	2,241,347	292	—	—

Here N_{raw} is the ambient number of coefficient variables before the finite-group decomposition, whereas B_{max} is the actual number of unknowns in the largest single modular-rank problem. The largest individual Wronskian layers have respectively 12,257, 8,848, and 12,184 unknowns. Thus none of the endpoint calculations forms one matrix with hundreds of thousands of columns: the complete frontier consists of 292 independent symmetry shards.

If B denotes the column dimension of one shard, the dense envelope for exact Gaussian elimination is $O(B^2)$ field elements of storage and $O(B^3)$ field operations. The implemented sparse elimination is usually smaller, but fill-in makes this dense envelope the safer planning model. For orientation, the completed cubic case $(m, t) = (14, 3)$ has $B_{\text{max}} = 8,830$. Its slowest block took 2,423 seconds in the historical endpoint run and 2,920 seconds in the later complete-profile rerun; monitoring of the historical run observed a 19.33 GiB working set on the recorded laptop. Using the completed cases in each family and a cubic-time dimensional extrapolation gives approximately 32–39 hours for the largest cubic shard, 40–50 hours for the largest conic–quintic shard, and 1.4 hours for the largest line–octic shard. These are deliberately rough resource estimates, not measured frontier running times or mathematical inputs; the sparsity and fill patterns differ among the three families. Nor does the sampled historical working set determine a per-shard peak. The matrix-free estimator does not construct the target rows, so exact row counts, nonzero counts, fill-in, and peak memory for the unrun frontier blocks are not yet known. We therefore make no claim here about a minimum hardware configuration; those quantities will be recorded with the completed certificates.

A cluster implementation remains straightforward. Each job is indexed by the degree pair, (m, t) , prime, character orbit, and, when present, involution eigenvalue. It generates its columns on demand, performs exact finite-field elimination, and writes an independent rank certificate; no communication with another shard is required. On each node, replacing tree-based sparse vectors by sorted packed rows and switching to blocked dense updates after fill-in should substantially reduce both memory traffic and overhead, while preserving the same exact modular proof. For scale, a square array for the largest block in each family occupies respectively 3.80, 1.07, and 2.28 GiB with 32-bit field entries, before row-space, polynomial-generation, fill-in, and elimination workspace. These figures are dimensional

baselines, not peak-memory predictions. The existing C++ interface already exposes the orbit and eigenvalue shards, so the 292 primary-prime jobs can be dispatched directly by an ordinary cluster scheduler. Full column rank at one good prime is logically sufficient; selected large blocks should also be replayed at a second prime as an implementation check.

Consequently, the limiting cases form a concrete, communication-free target for parallel C++ computation, but this feasibility analysis is neither a hardware guarantee nor a proof of any unperformed vanishing. If every listed normalized integral block has full column rank, the same argument gives the conditional constants above. As the blocks are completed, their row and nonzero counts, exact ranks, hashes, peak-memory records, and independently replayable certificates will be posted on the authors' evidence page.²

AUTHOR CONTRIBUTIONS

Song-Yan Xie formulated the problem investigated in this paper. Lei Hou and Song-Yan Xie developed the $\mathcal{O}(3)$ slanted-vector-field argument, including the new fields required in the two-component setting. The theoretical foundation of the Key Vanishing Lemma is the exponent-bound and finite-linear-algebra method developed by Hou and Xie in the preceding two papers and adapted here to two components. For the present coupled verification, Pengchao Wang introduced a new algorithmic idea and used it to substantially improve the exact finite algorithm. This conceptual improvement, rather than a mere code implementation, is what makes the required verification feasible on a standard laptop. All authors checked the mathematical arguments and computations and jointly revised and polished the manuscript.

AI-ASSISTED PREPARATION

Generative artificial-intelligence tools were used for language polishing. Under detailed author direction, they also assisted in drafting portions of the $\mathcal{O}(3)$ argument; the authors supplied the core ideas, precise mathematical constraints, and iterative guidance. Apart from that author-directed drafting assistance, the ideas and arguments are due to the authors. The authors independently verified every argument in its final form and accept full responsibility for the manuscript.

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²Updates and reproducible certificates will be made available at <https://xiesongyan.github.io/>.

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